MP2984

36V, Synchronous Buck-Boost Controller with Current Monitoring, I²C Interface, and Adjustable OCP via IPWM

DESCRIPTION

The MP2984 is a high-efficiency, synchronous, quad-switch, buck-boost controller that can regulate different output voltages across a wide input voltage (V_{IN}) range. The output voltage (V_{OUT}), V_{OUT} slew-rate, and output constant-current limit can be configured via the I²C interface. The MP2984 is suitable for USB power delivery (PD) applications in USB Type-C power supplies.

The MP2984 uses valley current control in buck mode and peak current control in boost mode to provide fast load transient response and smooth buck-boost mode transient. Forced continuous conduction mode (FCCM) and a configurable current limit support flexible design for various applications.

Full protection features include configurable over-current protection (OCP), over-voltage protection (OVP), and V_{IN} under-voltage lockout (UVLO) hysteresis.

The MP2984 is available in a QFN-32 (4mmx4mm) package.

FEATURES

- Wide 5V to 36V Operating V_{IN} Range
- Wide 6V to 36V Start-Up Input Voltage (V_{IN}) Range
- Flexible Control via the I²C Interface:
  - 0.5V to 28V Output Voltage (V_{OUT}) Range
  - 0.3V to 2.047V Reference Voltage (V_{REF}) Range with 1mV Step
  - Selectable V_{OUT} Slew Rate
  - Configurable Constant-Current Limit
- Adjustable Current Limit with <50mA Steps via IPWM
- Output Current Monitoring (IMON)
- Configurable Soft-Start Time (t_{SS})
- Switching Frequency (f_{SW}) with Frequency Spread Spectrum for Reduced EMI
- Integrated V_{OUT} Discharge
- 200kHz, 300kHz, 400kHz, or 600kHz Selectable f_{SW}
- Forced Continuous Conduction Mode (FCCM)
- Configurable V_{IN} Under-Voltage Lockout (UVLO) Hysteresis
- Minimum V_{IN} Regulation
- Over-Current Protection (OCP), Short-Circuit Protection (SCP), and Over-Voltage Protection (OVP)
- OCP, OVP, OTP, and PNG Interrupt Indication
- Available in a QFN-32 (4mmx4mm) Package

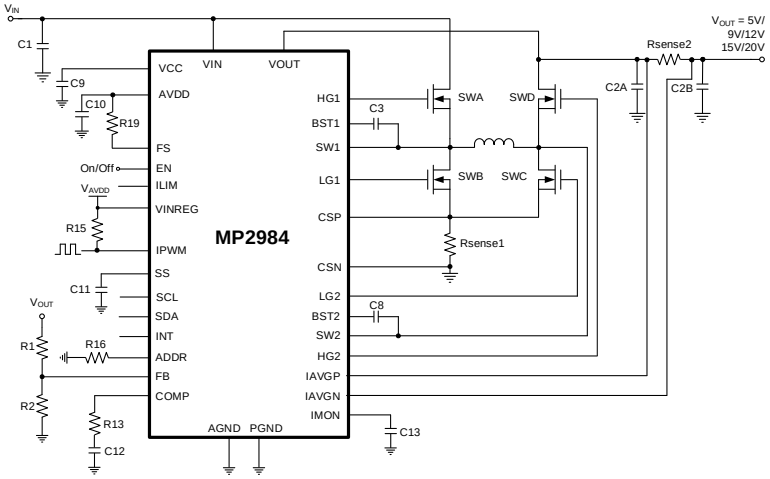
APPLICATIONS

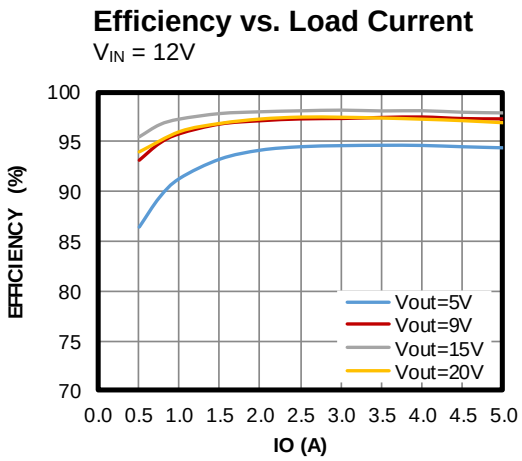
- USB Power Delivery (PD)
- Industrial PC Power Supplies

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TYPICAL APPLICATION





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ORDERING INFORMATION

Part Number	Package	Top Marking	MSL Rating
MP2984GR*	QFN-32 (4mmx4mm)	*See Below*	1
EVKT-MP2984	Evaluation kit		

* For Tape & Reel, add suffix -Z (e.g. MP2984GR-Z).

TOP MARKING

MPSYWW
MP2984
LLLLLL

MPS: MPS prefix
Y: Year code
WW: Week code
MP2984: Part number
LLLLLL: Lot number

EVALUATION KIT EVKT-MP2984

EVKT-MP2984 kit contents (items below can be ordered separately).

#	Part Number	Item	Quantity
1	EV2984-R-00B	MP2984GR evaluation board	1
2	EVKT-USBI2C-02-BAG	Includes USB to I²C communication interface device, USB cable, and ribbon cable	1

Order direct from MonolithicPower.com or our distributors.

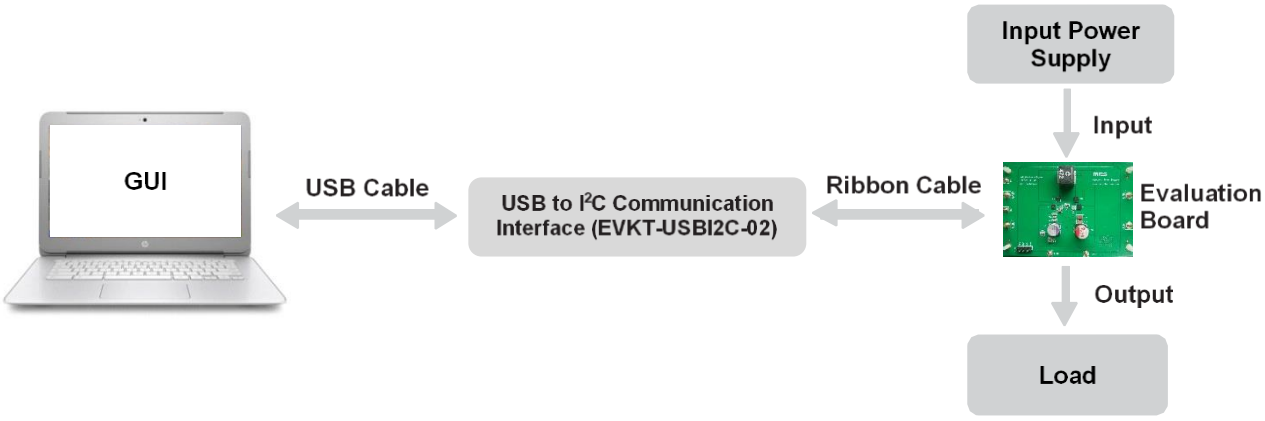
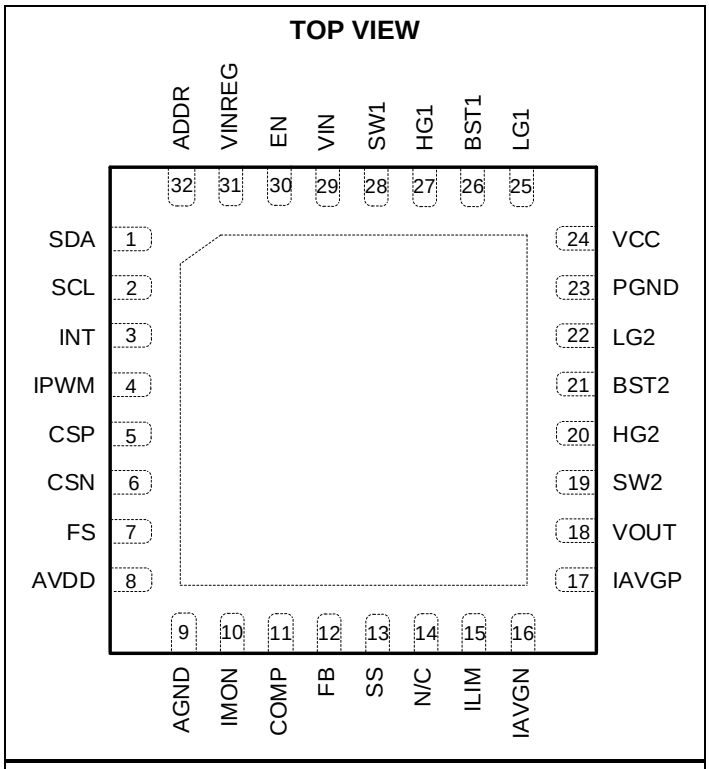


Figure 1: EVKT-MP2984 Evaluation Kit Set-Up

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PACKAGE REFERENCE



QFN-32 (4mmx4mm)

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PIN FUNCTIONS

Pin #	Name	Description
1	SDA	**I²C data signal.**
2	SCL	**I²C clock signal.**
3	INT	**Interrupt for over-current (OC), over-temperature (OT), over-voltage (OV), and PNG fault events.** If a PNG event occurs in a default set-up, the INT pin is masked off. INT is an open-drain output. If an interrupt event occurs, INT is pulled low. INT recovers to its initial status once the fault is removed. If the IC is disabled, then INT is an open drain.
4	IPWM	**Current dimming pulse-width modulation (PWM) signal input.** If the output current limit (I_{OUT_LIMIT}) is configured via the I²C register, then the IPWM pin input signal duty cycle controls the average I_{OUT_LIMIT}. Pull IPWM up to the AVDD pin internally via a 1MΩ resistor to set a 100% duty high-level voltage. It is recommended to connect an additional 100kΩ external pull-up resistor to IPWM.
5	CSP	**Positive switching current-sense input.** Connect the CSP pin to the high side (HS) of the current-sense resistor.
6	CSN	**Negative switching current-sense input.** Connect the CSN pin to the low side (LS) of the current-sense resistor.
7	FS	**Switching frequency (f_{SW}) bits default value setting.** During start-up, the two voltage levels on the FS pin set the default f_{SW} values. The first FS voltage level is 0.51 x V_{AVDD} to 0.68 x V_{AVDD}, and the FSW bits are set to 00 (200kHz default f_{SW}). The second FS voltage level is 0.74 x V_{AVDD} or greater, and the FSW bits are set to 10 (400kHz default f_{SW}). Do not float FS or connect FS to GND. For different voltage level settings, see the Electrical Characteristics section on page 9. Changing the FS voltage level after start-up does not change the FSW bit settings. After start-up, the FSW bits can be set via the I²C interface.
8	AVDD	**Internal control circuit bias supply.** Use a ≥2.2μF decoupling capacitor to decouple the AVDD pin.
9	AGND	**Analog ground.**
10	IMON	**Current monitoring output.** The IMON pin outputs the voltage signal between the IAVGP and IAVGN pins. The MP2984 senses the average load current via a current-sense resistor to output this signal.
11	COMP	**Internal error amplifier (EA) output.** For loop compensation, connect a capacitor and a resistor in series between the COMP and AGND pins.
12	FB	**Output voltage (V_{OUT}) feedback.** Connect a resistor divider between the VOUT and FB pins.
13	SS	**Soft start (SS) setting.** Connect an external capacitor to the SS pin. SS also sets the hiccup mode off time.
14	NC	**No connection.**
15	ILIM	**ILIM bits default value setting.** During start-up, the two voltage levels on the ILIM pin set the default current limit (I_{LIMIT}) values. At the first ILIM voltage level, the ILIM bits are set to 001 (32mV default I_{LIMIT}). At the second ILIM voltage level, the ILIM bits are set to 011 (45mV default I_{LIMIT}). At the third ILIM voltage level, the ILIM bits are set to 101 (56mV default I_{LIMIT}). At the fourth ILIM voltage level, the ILIM bits are set to 111 (68mV default I_{LIMIT}). In default set-up, float the ILIM pin to set the ILIMT bits to 111. For different voltage level settings, see the Electrical Characteristics section on page 9. Changing the ILIM voltage level after start-up does not change the ILIM bit settings. After start-up, the ILIM bits can be set via the I²C interface.
16	IAVGN	**Negative average current limit sense input.** Connect the IAVGN and IAVGP pins to the output rail's positive terminal to use IAVGN and IAVGP to set I_{OUT_LIMIT}.
17	IAVGP	**Positive average current limit sense input.** Connect the IAVGN and IAVGP pins to the output rail's positive terminal to use IAVGN and IAVGP to set I_{OUT_LIMIT}.
18	VOUT	**V_{OUT} voltage sense input.** The VOUT pin supplies power to the VCC pin based on the VCC power logic. Connect VOUT to the output capacitor (C_{OUT}).
19	SW2	**Converter boost switch node.** Connect the SW2 pin to SWD's source and SWC's drain.

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PIN FUNCTIONS *(continued)*

Pin #	Name	Description
20	HG2	**Boost high-side MOSFET (HS-FET) gate driver.** Connect the HG2 pin to SWD's gate.
21	BST2	**Boost HS-FET gate driver bootstrap power.** Connect a capacitor between the BST2 and SW2 pins. VCC or BST1 can supply BST2.
22	LG2	**Boost low-side MOSFET (LS-FET) gate driver.** Connect the LG2 pin to SWC's gate.
23	PGND	**Power ground.** Gate driver current return pin.
24	VCC	**Driver circuit and internal bias supply**. Use a ≥2.2μF ceramic decoupling capacitor to decouple VCC. Place this capacitor as close to VCC as possible. VOUT or VIN can supply VCC.
25	LG1	**Buck LS-FET gate driver pin.** Connect LG1 to SWB's gate.
26	BST1	**Buck HS-FET gate driver bootstrap power.** Connect a capacitor between the BST1 and SW1 pins. VCC or BST2 can supply BST1.
27	HG1	**Buck HS-FET gate driver.** Connect directly to the gate of SWA.
28	SW1	**Converter buck switch node.** Connect to the source of SWA and the drain of SWB.
29	VIN	**Power supply and voltage sense input.**
30	EN	**Enable.** If not used, connect the EN pin to the input source for automatic start-up. EN can also configure V_{IN} under-voltage lockout (UVLO) protection. Do not float EN.
31	VINREG	**V_{IN} regulation.** The VINREG pin sets the minimum operating V_{IN} during switching. Connect VINREG to AVDD if not used.
32	ADDR	**I²C slave address setting.** The ADDR pin sets the ENPWR bit's default value.

ABSOLUTE MAXIMUM RATINGS (1)

VIN, EN ..... -0.3V to +40V
VOUT, IAVGP, IAVGN ..... -0.3V to +30V
VCC ..... -0.3V to +8.5V
SW1, SW2 .....
.....-1V to +40V (-5V to +45V for <20ns)
LG1, LG2 .....
.....-0.3V to +10V (-2V to +11V for <20ns)
BST1, HG1 ..... -0.3V to V_{SW1} + 8.5V
BST2, HG2 ..... -0.3V to V_{SW2} + 8.5V
All other pins ..... -0.3V to +6.5V
Continuous power dissipation (2) (5) 4.4W
Junction temperature ..... 150°C
Lead temperature ..... 260°C
Storage temperature ..... -65°C to +150°C

Recommended Operating Conditions (3)

Start-up voltage (V_{SU})..... 6V to 36V
Operating input voltage (V_{IN}) (4) ..... 5V to 36V
Output voltage (V_{OUT})..... 0.5V to 28V
Operating junction temp (T_J) .... -40°C to +125°C

Thermal Resistance θ_{JA} θ_{JC}

EV2984-R-00B (5)	.....	28	8 °C/W
JESD51-7 (6)	.....	42	9 °C/W

Notes:

1) Exceeding these ratings may damage the device.
2) The maximum allowable power dissipation is a function of the maximum junction temperature T_J (MAX), the junction-to-ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by P_D (MAX) = (T_J (MAX) - T_A) / θ_{JA}. Exceeding the maximum allowable power dissipation can cause excessive die temperature, and the device may go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.
3) The device is not guaranteed to function outside of its operating conditions.
4) This value is the operating V_{IN} after V_{OUT} has been regulated at ≥5V, and the VCC load is < 10mA.
5) Measured on the MPS evaluation board EV2984-R-00B, 91.4mmx66mm, 6-layer PCB, 2oz-1oz-1oz-1oz-1oz-2oz.
6) The value of θ_{JA} given in this table is only valid for comparison with other packages, and cannot be used for design purposes. These values were calculated in accordance with JESD51-7, and simulated on a specified JEDEC board. They do not represent the performance obtained in an actual application.

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ELECTRICAL CHARACTERISTICS

V_{IN} = 12V, V_{OUT} = 12V, V_{EN} = 2V, T_J = -40°C to 125°C [7], typical values are tested at T_J = 25°C, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Power Supply						
Operating VCC voltage	V_{CC}	V_{IN} = 6V or V_{OUT} = 6V, 0mA to 20mA on VCC	5.1	5.95	6	V
		V_{IN} = 12V or V_{OUT} = 12V, 0mA to 60mA on VCC	6.7	7.2	7.7	V
VIN under-voltage lockout (UVLO) rising threshold [8]	$V_{IN_UVLO_RISING}$	V_{IN} rising	5	5.5	5.9	V
VCC UVLO falling threshold [8]	$V_{CC_UVLO_FALLING}$	V_{CC} falling	3.8	4.3	4.8	V
VCC power source threshold	V_{IN_VCC}	V_{OUT} = 12V, V_{IN} ramps up from 5V to 10V	8.1	8.8	9.5	V
	V_{OUT_VCC}	V_{IN} = 12V, V_{OUT} ramps up from 5V to 10V	8.1	8.8	9.5	V
AVDD voltage	V_{AVDD}	V_{IN} = 12V, 0mA to 5mA	4.7	5.2	5.6	V
Shutdown current	I_{SD}	Measured on the VIN and VOUT pins, V_{EN} = 0V			5	µA
		ENPWR bit = 0, V_{IN} = 12V, V_{OUT} = 0V, measured on the VIN pin, V_{EN} = 2V	300	450	600	µA
Enable (EN) Control						
EN start-up threshold voltage	V_{EN_SU}	V_{EN} rising, switching	1.25	1.35	1.45	V
V_{EN} high threshold	V_{EN_HIGH}	V_{EN} rising, micro-power			1.1	V
V_{EN} low threshold	V_{EN_LOW}	V_{EN} falling, micro-power	0.4			V
EN start-up hysteresis current	I_{EN_HYS}	V_{EN} > V_{EN_SU}, EN source current	3.2	4.7	6.2	µA
EN current	I_{EN}	V_{EN} = 0V, 3.3V		0.01		µA
ENPWR start-up delay [9]	t_{DELAY_ENPWR}	From ENPWR = 1 to switching, C_{SS} = 47nF		1		ms
Feedback (FB) Control						
FB reference voltage	V_{REF}	VREF bits = 7FFH, T_J = 25°C	-1	2.047	+1	% of V_{REF}
		VREF bits = 7FFH, T_J = -40°C to 125°C	-2	2.047	+2	
		VREF bits = 1F4H, T_J = 25°C	-2	0.5	+2	
		VREF bits = 1F4H, T_J = -40°C to 125°C	-3	0.5	+3	
FB current	I_{FB}	V_{FB} = 0.52V			200	nA
Error amplifier (EA) transconductance	G_{EA}	V_{FB} = V_{REF} + 10mV, V_{COMP} = 2.5V		1220		µA/V
COMP to Current Sense Gain [9]	G_{CS}	ΔV_{CS} / ΔV_{COMP}		200		mV/V
SS Charge Current	I_{CHG_SS}	During soft start and overload recovery	2	6	10	µA
SS Discharge Current	I_{DSG_SS}	After hiccup protection is triggered		1		µA
V_{REF} slew rate	t_{SR}	SR = 00	25	38	51	mV/ ms
		SR = 11	130	150	170	mV/ ms
VINREG voltage	V_{VINREG}	T_J = 25°C	1.188	1.2	1.212	V
		T_J = -40°C to 125°C	1.182	1.2	1.218	V
VINREG current	I_{VINREG}	V_{VINREG} = 1.25V			50	nA
Current Limit						
Buck valley current limit	I_{LIMIT_VALLEY}		113	133	153	mV

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ELECTRICAL CHARACTERISTICS *(continued)*

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Parameter	Symbol	Condition	Min	Typ	Max	Units
Boost peak current limit	I_{LIMIT_PEAK}		130	150	170	mV
Over-current protection (OCP) threshold [9]	V_{OCP}			60		% of V_{REF}
Constant-current limit	I_{LIMIT_CC}	ILIM bits = 011, V_{IAVGN} = 12V, V_{IAVGP} ramps up	40	45	50	mV
		ILIM bits = 111, V_{IAVGN} = 12V, V_{IAVGP} ramps up	62.5	68	73.5	mV
		ILIM bits = 111, V_{IAVGN} = 12V, T_J = 25°C, V_{IAVGP} ramps up	-5%	68	+5%	mV
IPWM input high threshold	V_{IPWM_HIGH}				1.2	V
IPWM input low threshold	V_{IPWM_LOW}		0.4			V
IPWM to AVDD internal pull-up resistor	R_{IPWM}			1		MΩ
Current limit dimming	$I_{DIMMING}$	48% IPWM duty, 20kHz signal, T_J = 25°C, ILIM bits = 111, measure load current limit	33	36	41	mV
		IPWM duty = 71.5% 20kHz signal, T_J = 25°C, ILIM bits = 111, measure load current limit	48	51.5	57	mV
CSP and CSN current bias	I_{CS_BIAS}	V_{CSP} = V_{CSN} = 0V		70		µA
IAVGP and IAVGN current bias	I_{AV_BIAS}	IAVGN = 5V, IAVGN = 20V, V_{IAVGP} - V_{IAVGN} = 40mV		55		µA
Switching Frequency						
Switching frequency	f_{SW}	FSW bits = 10, V_{OUT} = 5V	320	400	480	kHz
		FSW bits = 00, V_{OUT} = 5V	140	200	260	kHz
Frequency spread span [9]	f_{SS}	Dither bit = 1		±6		% of f_{SW}
Frequency spread spectrum (FSS) modulation frequency [9]	f_{SS_MOD}	Dither bit = 1		2		kHz
Gate Driver						
Gate source current capability [9]	I_{HG_SOURCE}	V_{CC} = 7.2V, 4.7nF load		0.7		A
	I_{LG_SOURCE}			0.85		A
Gate sink current capability [9]	I_{HG_SINK}	V_{CC} = 7.2V, 4.7nF load		1.6		A
	I_{LG_SINK}			2		A
Low-side (LS) gate output high voltage	V_{LS_HIGH}		V_{CC} - 0.05			V
LS gate output low voltage	V_{LS_LOW}				0.05	V
High-side (HS) gate output high voltage	V_{HS_HIGH}		(V_{BST} - V_{SW}) - 0.05			V
HS gate output low voltage	V_{HS_LOW}				0.05	V
Dead-time between HS gate and LS gate [9]	t_{DEAD}			30		ns

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 12V, V_{EN} = 2V, T_J = -40°C to 125°C [7], typical values are tested at T_J = 25°C, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Over-Voltage Protection (OVP)						
FB OVP rising threshold	V_{OVP_RISING}		119	127	135	% of V_{REF}
FB OVP falling threshold	$V_{OVP_FALLING}$		104	111	118	% of V_{REF}
Thermal Protection						
Thermal shutdown [9]	T_{SD}			150		°C
Thermal shutdown hysteresis [9]	T_{SD_HYS}			25		°C
Power Good (PG)						
PG high threshold	$V_{PG_FALLING_HIGH}$	PNG bit is set to 1, INT is pulled low	110	117	124	% of V_{REF}
	$V_{PG_RISING_HIGH}$	PNG bit is reset to 0, INT is pulled high	101	106.5	112	
PG low threshold	$V_{PG_FALLING_LOW}$	PNG bit set to 1, INT is pulled low	80	85.5	91	% of V_{REF}
	$V_{PG_RISING_LOW}$	PNG bit resets to 0, INT is pulled high	85	91	97	
PG delay (INT response to PNG fault)	t_{DELAY_PG}	Low to high		10		µs
		V_{OUT} UVLO, high to low		2		µs
		V_{OUT} OVP, high to low		6.5		ms
INT sink current capability	I_{SINK_INT}	4mA sink		0.1	0.4	V
INT leakage current	I_{LKG_INT}	V_{INT} = 5V			1	µA
I²C Interface						
Input logic low voltage	V_{IN_LOW}	SCL, SDA			0.8	V
Input logic high voltage	V_{IN_HIGH}	SCL, SDA	2			V
Logic input current	$I_{SCL_SDA_LKG}$	SCL = 5V, SDA = 5V	-1		+1	µA
Output logic low voltage	V_{OUT_LOW}	SDA, 4mA sink			0.4	V
ADDR Setting and ILIM Voltage Setting						
Voltage level 1 setting	ADDR1	Sets I²C address 60H (1100 000)			0.23	V_{AVDD}
Voltage level 2 setting	ADDR2	Sets I²C address 62H (1100 010)	0.27		0.47	V_{AVDD}
Voltage level 3 setting	ADDR3	Sets I²C address 64H (1100 100)	0.51		0.68	V_{AVDD}
Voltage level 4 setting	ADDR4	Sets I²C address 66H (1100 110)	0.74			V_{AVDD}
Pin to GND pull-down resistance		ADDR, ILIM pins		2		MΩ
FS Pin Voltage Setting						
Voltage level 1 setting	V_{FS_LEVEL1}	Sets FSW bits to 00	0.51		0.68	V_{AVDD}
Voltage level 2 setting	V_{FS_LEVEL2}	Sets FSW bits to 10	0.74			V_{AVDD}
Current Monitoring						
IMON V_{OUT} gain	G_{IMON}	IAVG sense voltage = 5mV		18.8		V/V
		IAVG sense voltage = 55mV	16.92	18.8	20.68	V/V

Notes:

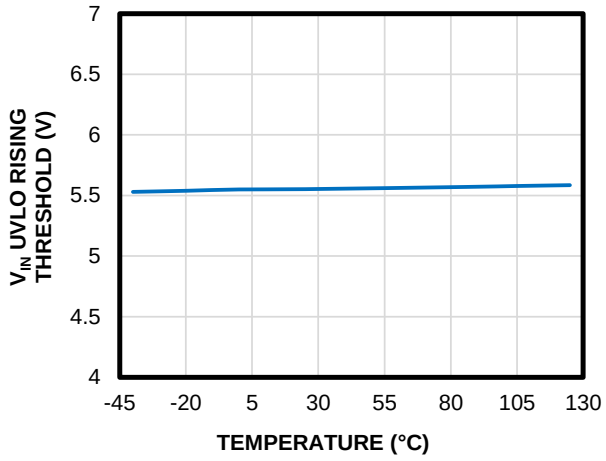
7) Not tested in production. Guaranteed by over-temperature (OT) correlation.
8) The MP2984 has a minimum start-up voltage of 6, and its V_{IN} UVLO falling threshold is lower than its VCC UVLO falling.
9) Guaranteed by engineer sample characterization.

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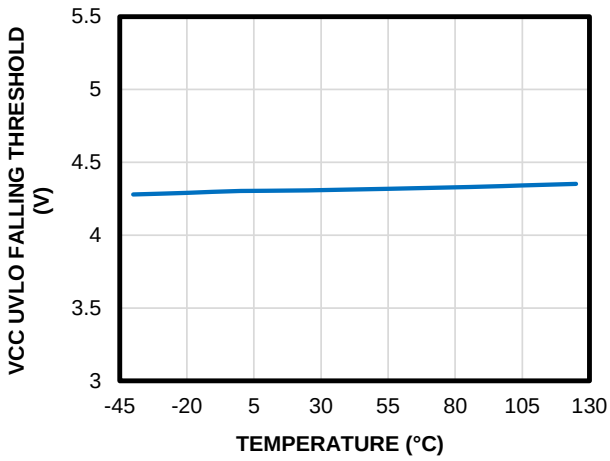
TYPICAL PERFORMANCE CHARACTERISTICS

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

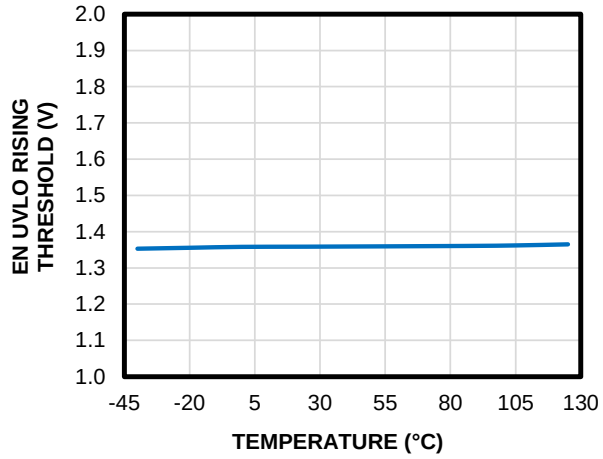
V_{IN} UVLO Rising vs. Temperature



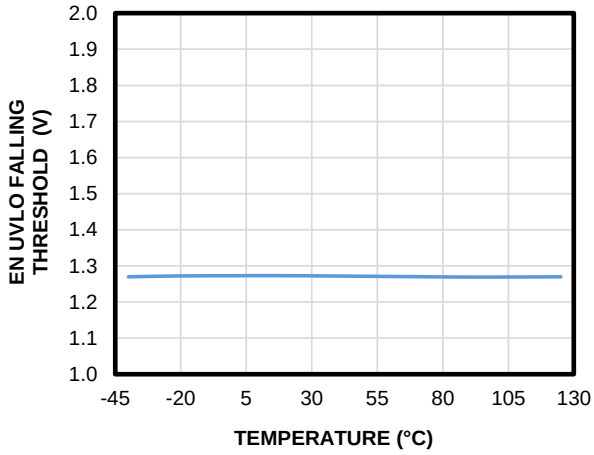
VCC UVLO Falling vs. Temperature



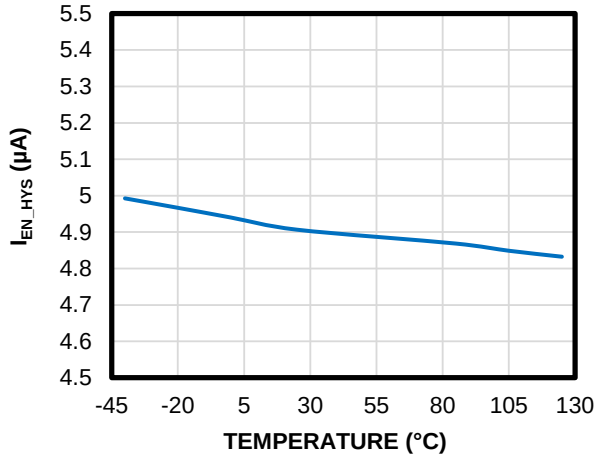
EN UVLO Rising Threshold vs. Temperature



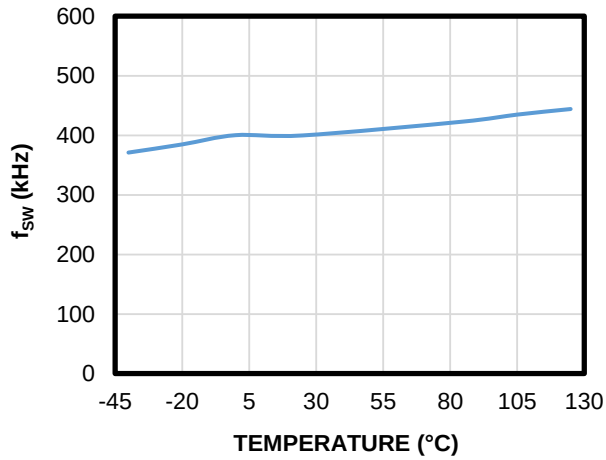
EN UVLO Falling Threshold vs. Temperature



EN Start-Up Hysteresis Current vs. Temperature



Switching Frequency vs. Temperature

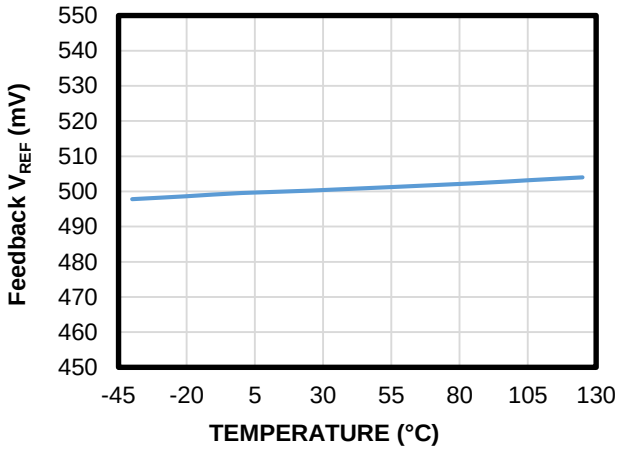


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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

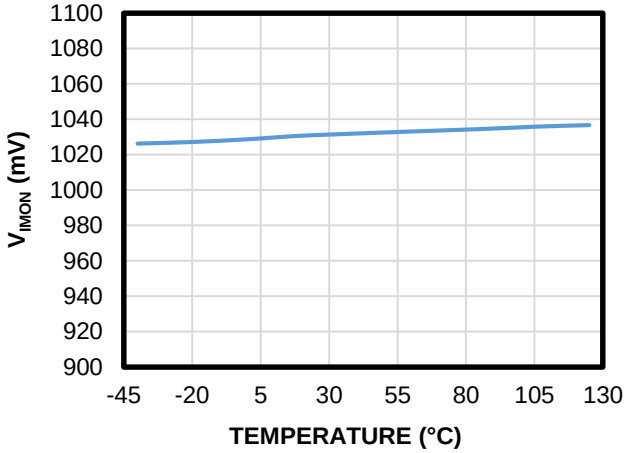
V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

Feedback Reference Voltage vs. Temperature

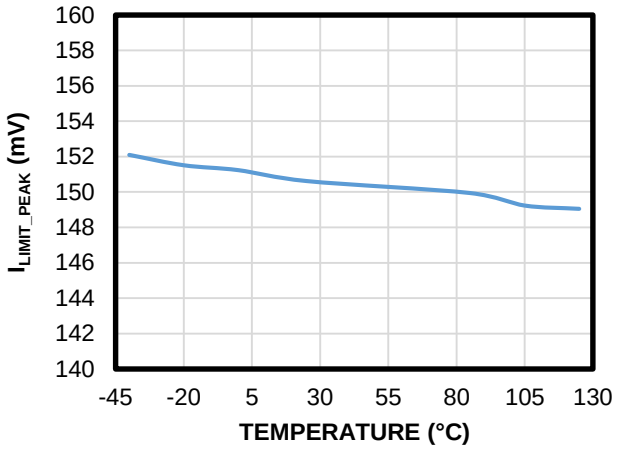


IMON Voltage vs. Temperature

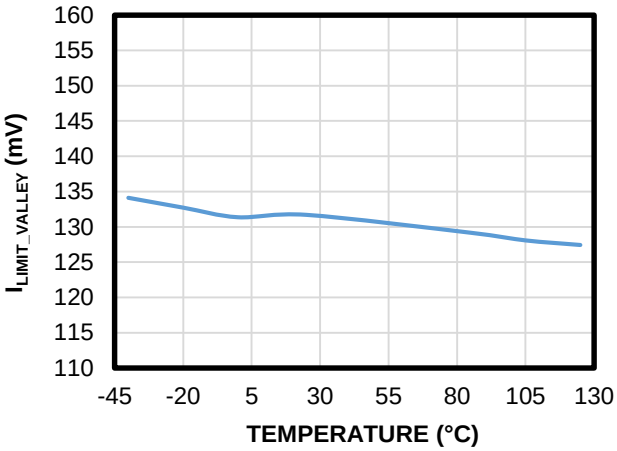
V_{IAVGP} - V_{IAVGN} = 55mV



Boost Peak Current Limit vs. Temperature

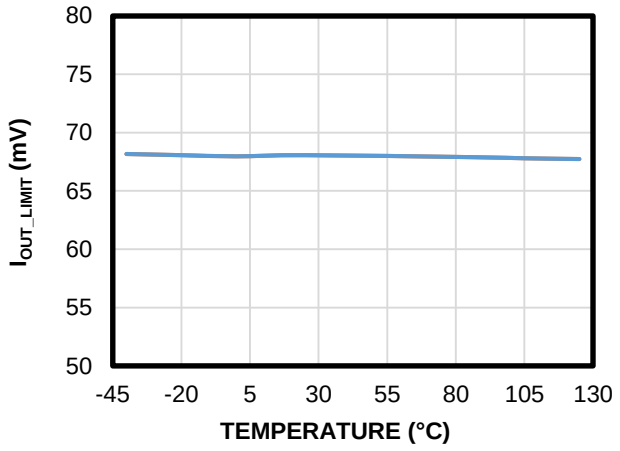


Buck Valley Current Limit vs. Temperature



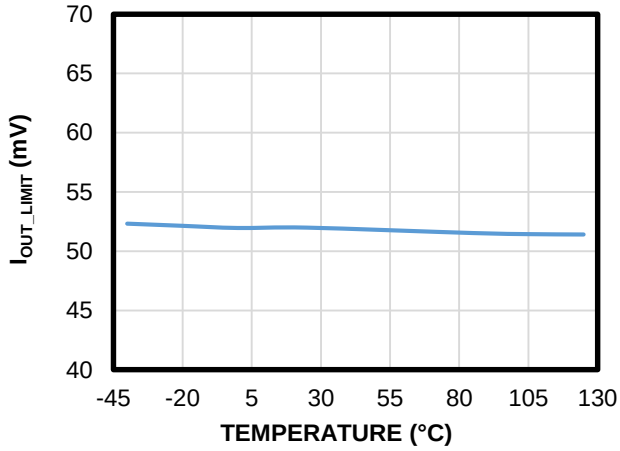
Output Current Limit vs. Temperature

ILIM bit = 111



Output Current Limit vs. Temperature

ILIM bit = 111, 71.5% duty cycle



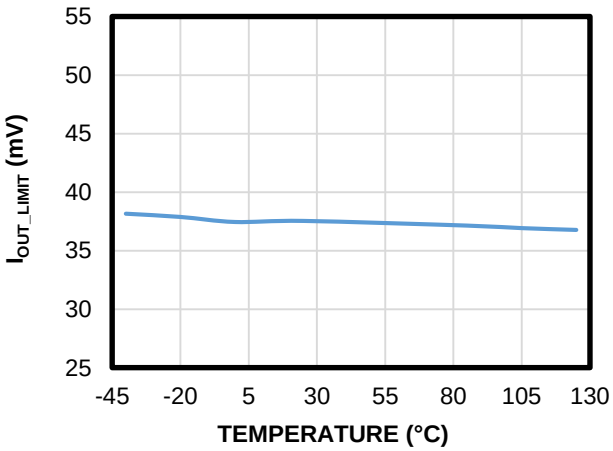
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

Output Current Limit vs. Temperature

ILIM bit = 111, 48% duty cycle



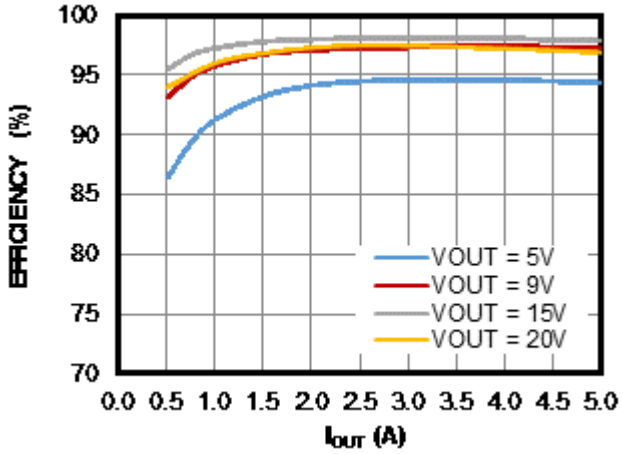
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

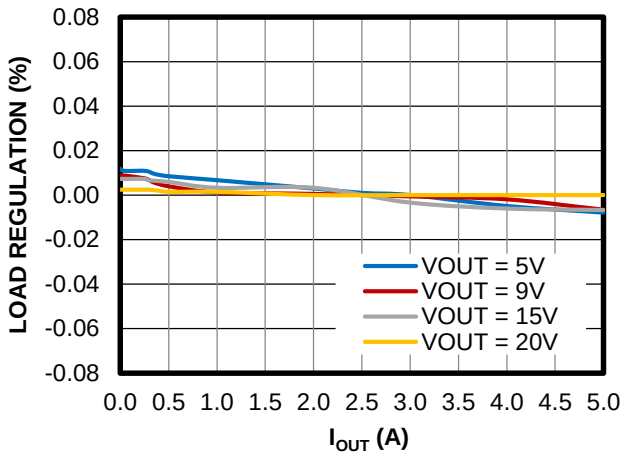
Efficiency vs. Output Current

V_{IN} = 12V



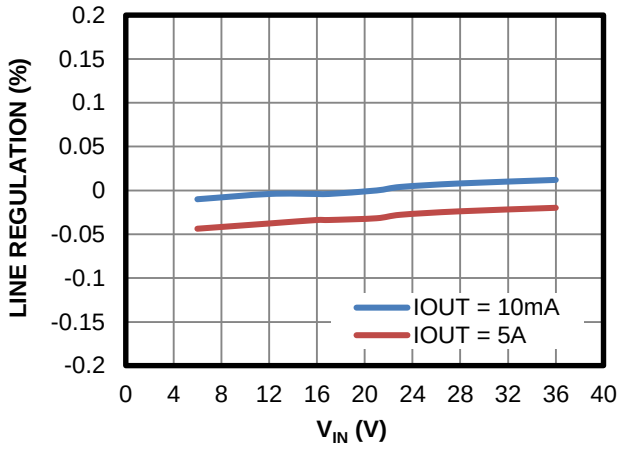
Load Regulation

V_{IN} = 12V



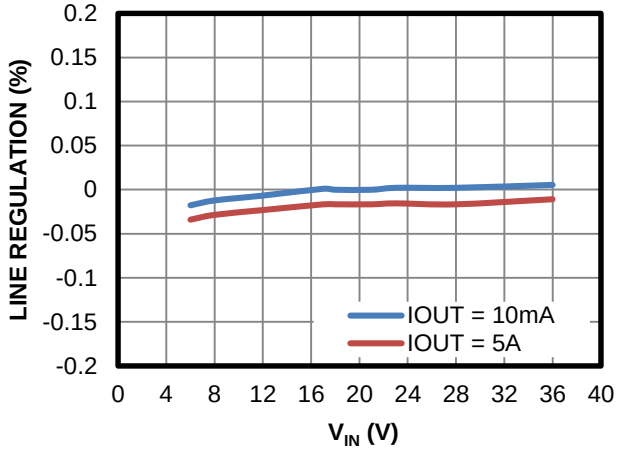
Line Regulation

V_{OUT} = 5V



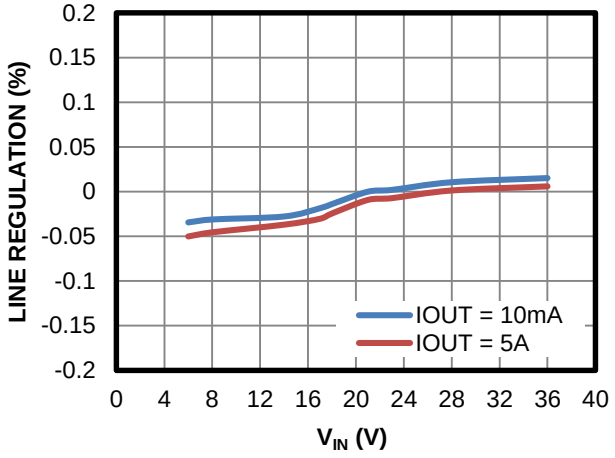
Line Regulation

V_{OUT} = 9V



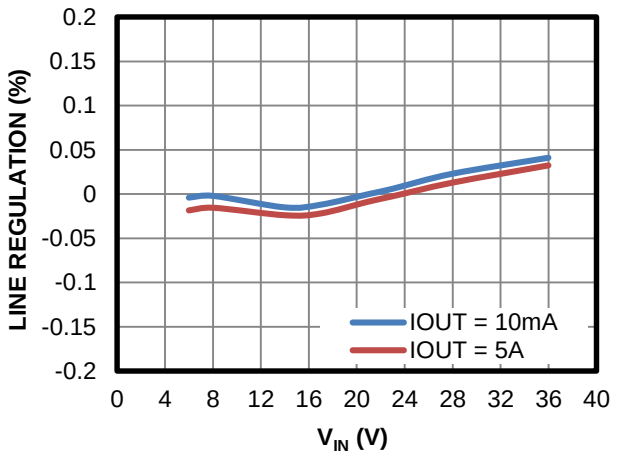
Line Regulation

V_{OUT} = 15V



Line Regulation

V_{OUT} = 20V



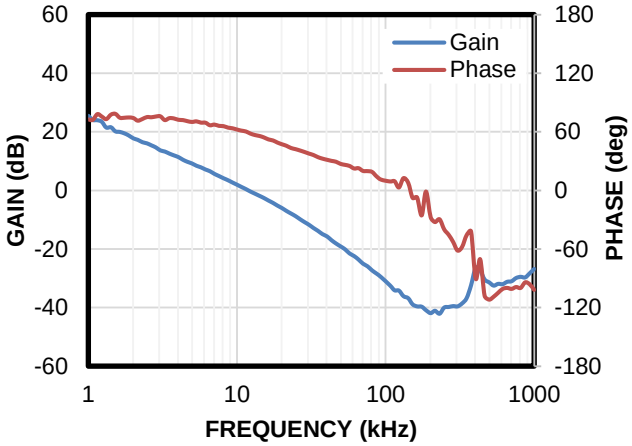
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

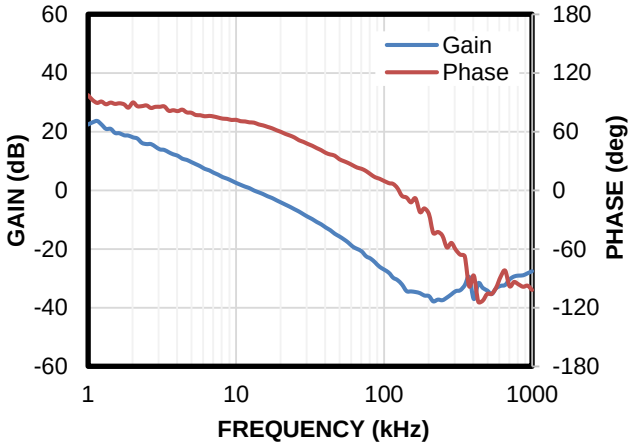
Bode Plot

V_{OUT} = 5V, I_{OUT} = 3A, BW = 12.37kHz, PM = 58.149°



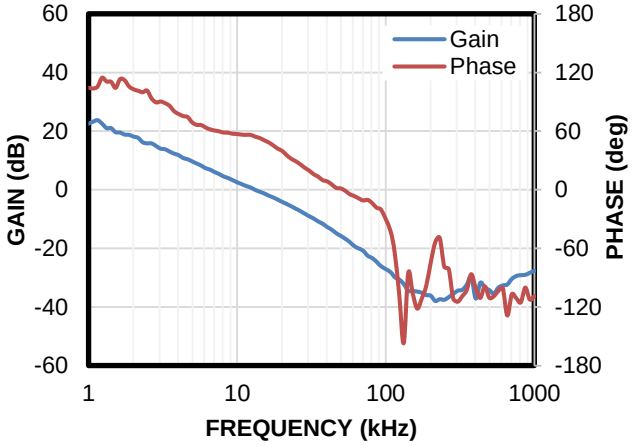
Bode Plot

V_{OUT} = 9V, I_{OUT} = 3A, BW = 13.81kHz, PM = 68.188°



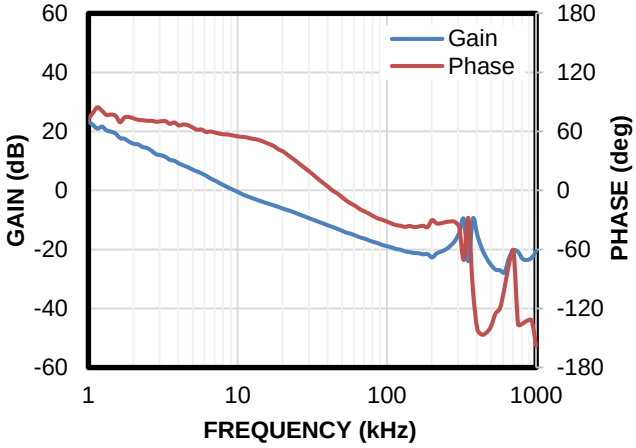
Bode Plot

V_{OUT} = 15V, I_{OUT} = 3A, BW = 9.16 kHz, PM =57.613 deg



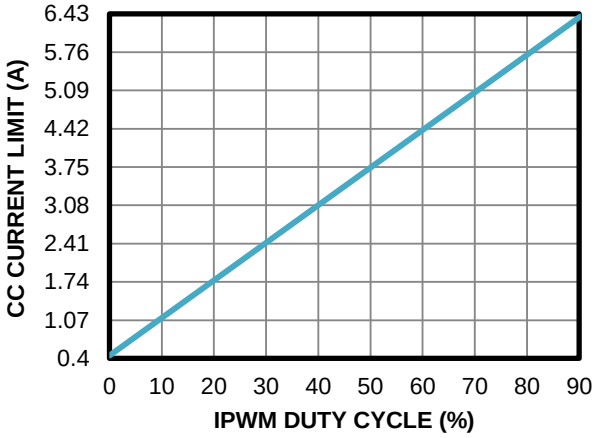
Bode Plot

V_{OUT} = 20V, I_{OUT} = 5A, BW = 10 kHz, PM = 54.907deg



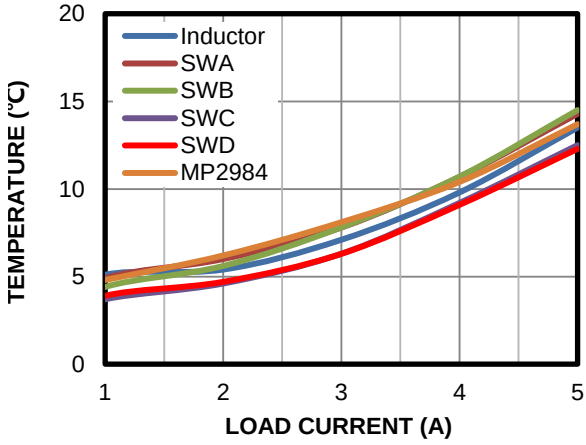
Constant-Current Limit vs. IPWM Duty Cycle

ILIM = 111, R_{SENSE2} = 10mΩ



Case Temperature Rise

V_{IN} = 12V, V_{OUT} = 5V, FSW bits = 10, tested on the EV2984-R-00B



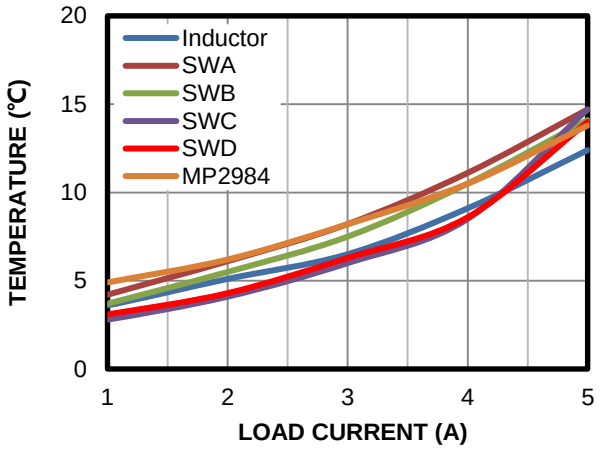
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

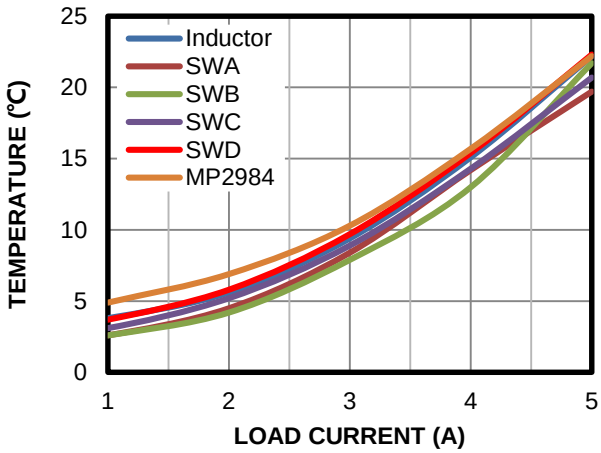
Case Temperature Rise

V_{IN} = 12V, V_{OUT} = 9V, FSW bits = 10, tested on the EV2984-R-00B



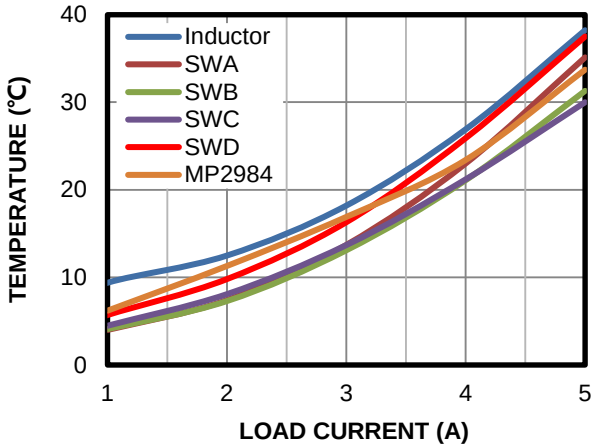
Case Temperature Rise

V_{IN} = 12V, V_{OUT} = 15V, FSW bits = 10, tested on the EV2984-R-00B



Case Temperature Rise

V_{IN} = 12V, V_{OUT} = 20V, FSW bits = 10, tested on the EV2984-R-00B

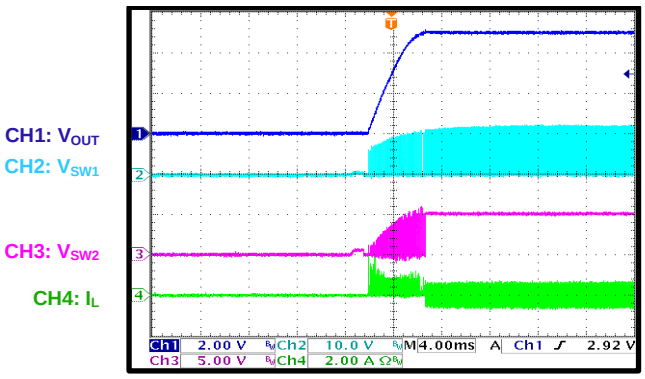


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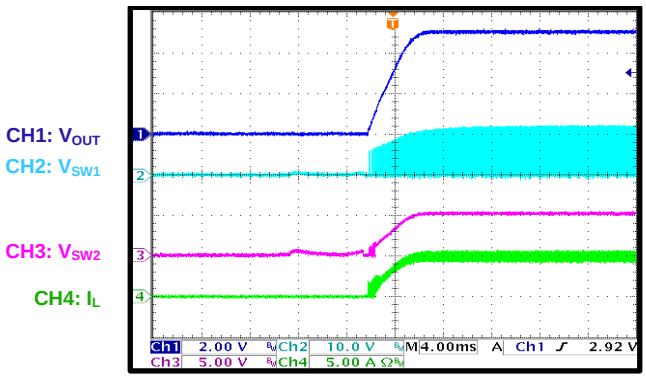
TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

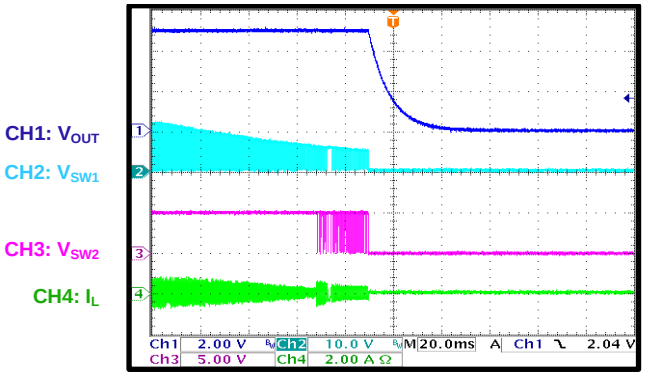
Start-Up through VIN
I_{LOAD} = 0A



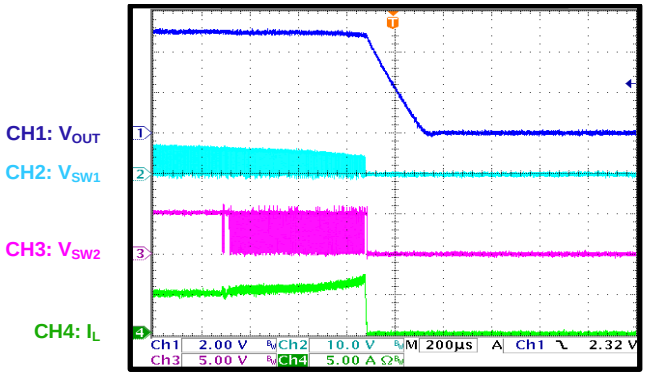
Start-Up through VIN
I_{LOAD} = 5A



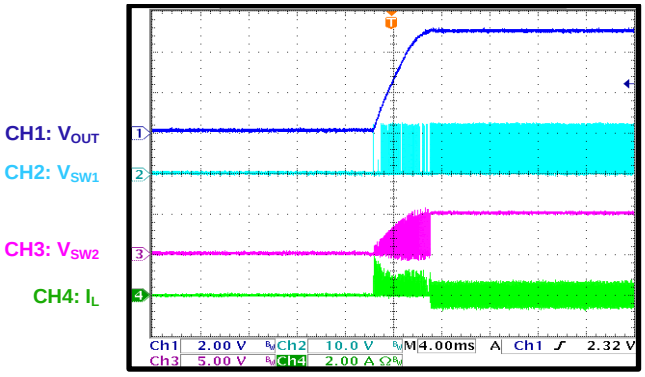
Shutdown through VIN
I_{LOAD} = 0A



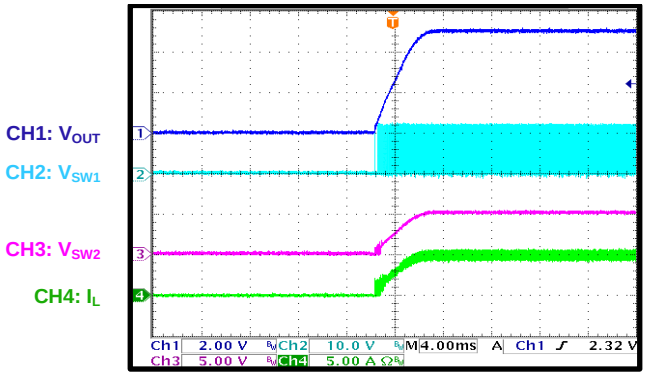
Shutdown through VIN
I_{LOAD} = 5A



ENPWR Enabled through the I²C
I_{LOAD} = 0A



ENPWR Enabled through the I²C
I_{LOAD} = 5A

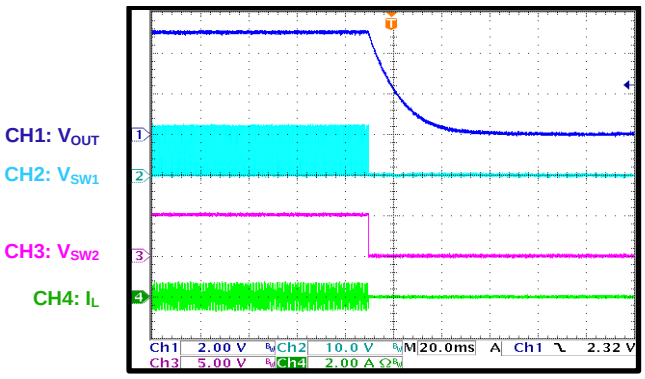


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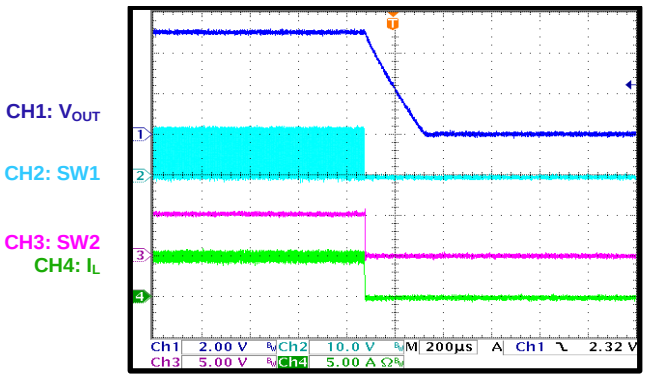
TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

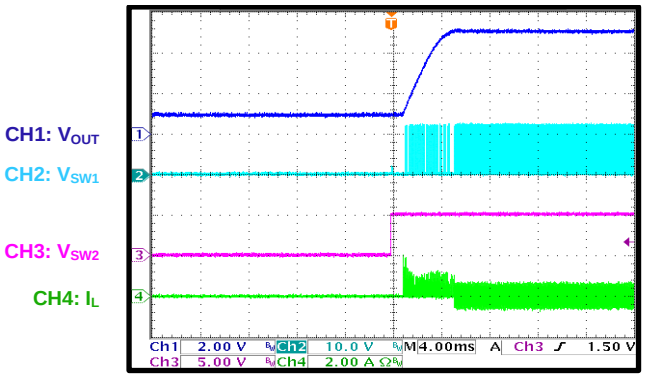
ENPWR Disabled through the I²C
I_{LOAD} = 0A



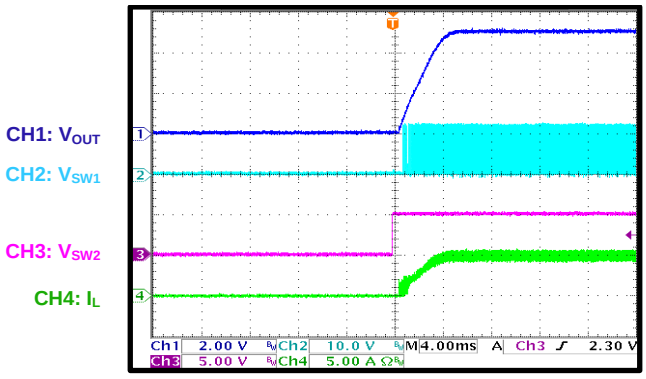
ENPWR Disabled through the I²C
I_{LOAD} = 5A



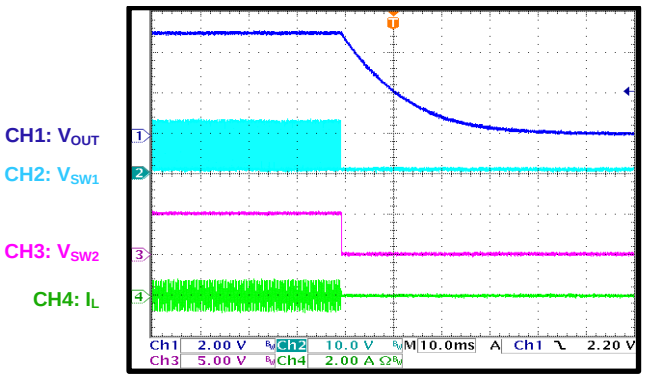
Start-Up through EN
I_{LOAD} = 0A



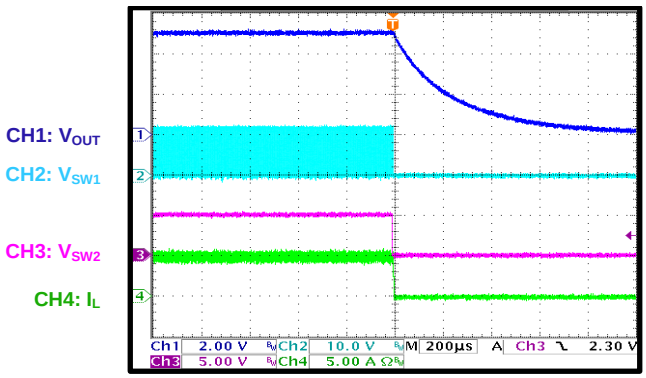
Start-Up through EN
I_{LOAD} = 5A



Shutdown through EN
I_{LOAD} = 0A



Shutdown through EN
I_{LOAD} = 5A

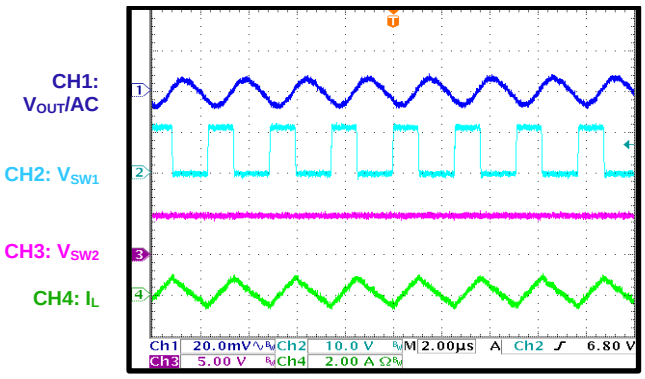


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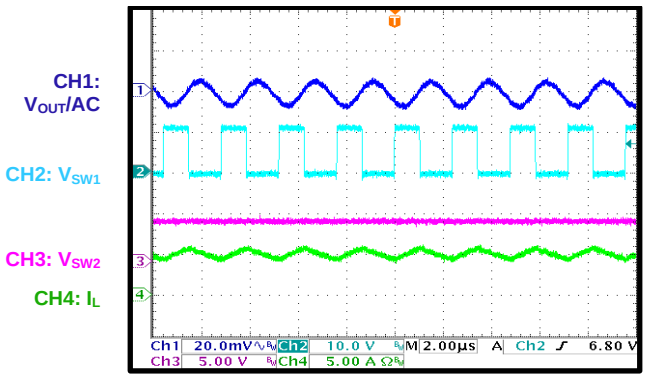
TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

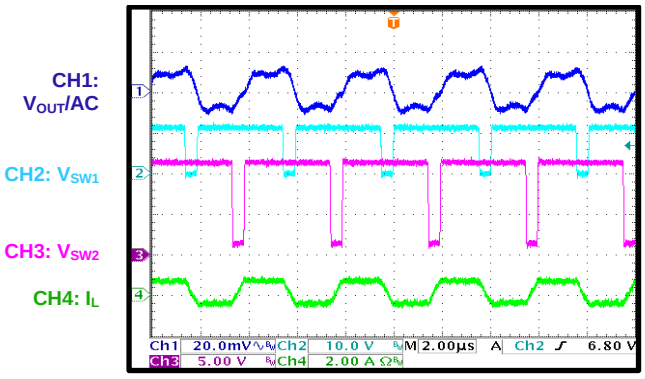
Steady State
V_{OUT} = 5V, I_{LOAD} = 0A



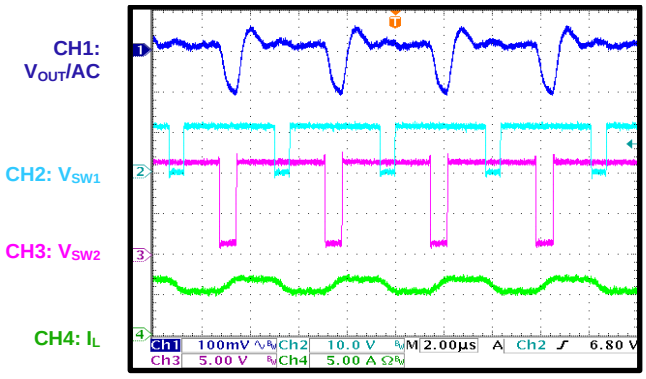
Steady State
V_{OUT} = 5V, I_{LOAD} = 5A



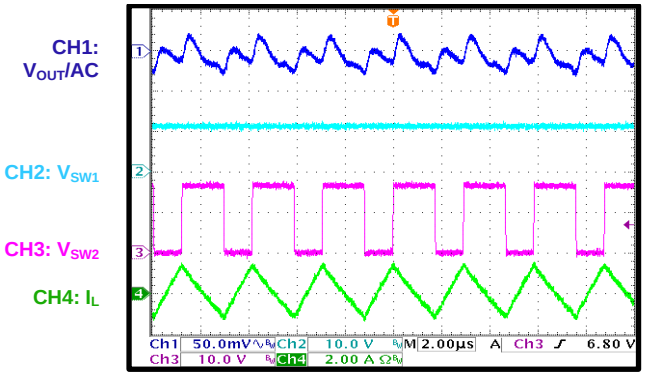
Steady State
V_{OUT} = 12V, I_{LOAD} = 0A



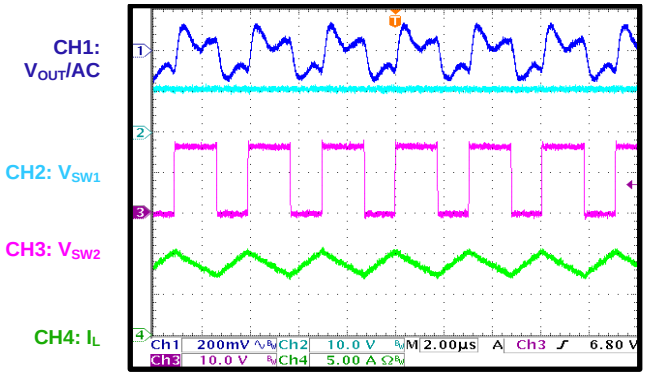
Steady State
V_{OUT} = 12V, I_{LOAD} = 5A



Steady State
V_{OUT} = 20V, I_{LOAD} = 0A



Steady State
V_{OUT} = 20V, I_{LOAD} = 5A



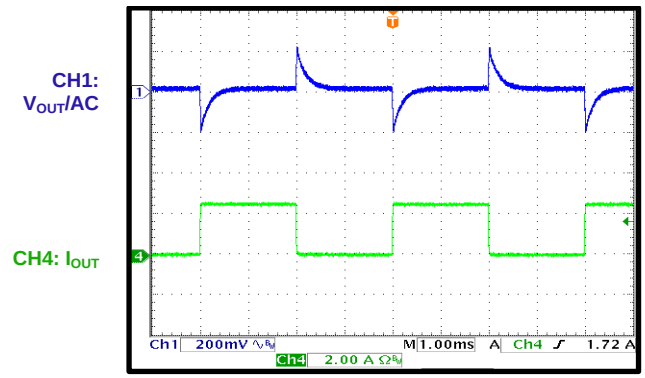
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

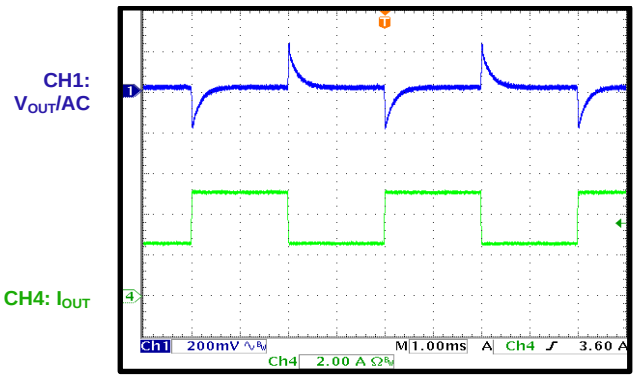
Load Transient

V_{IN} = 12V, V_{OUT} = 5V, I_{LOAD} = 0A to 2.5A, 150mA/µs



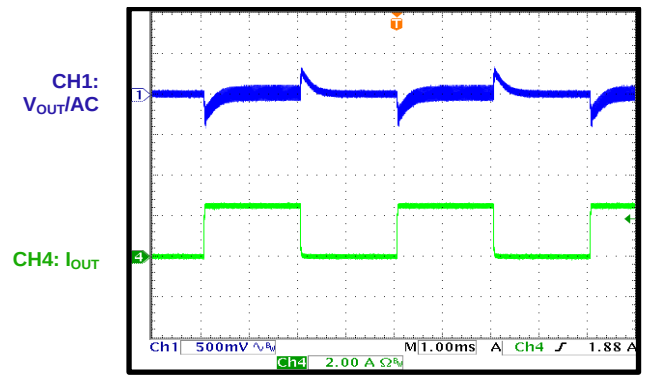
Load Transient

V_{IN} = 12V, V_{OUT} = 5V, I_{LOAD} = 2.5A to 5A, 150mA/µs



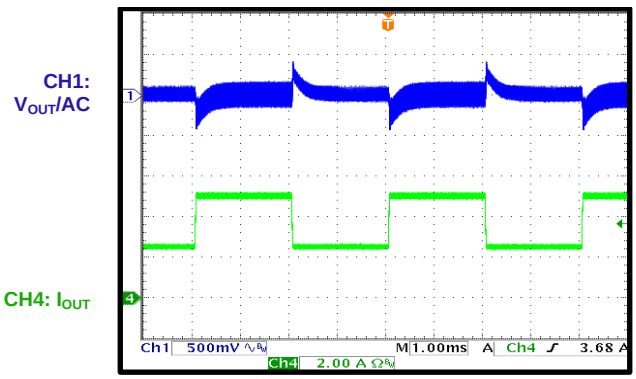
Load Transient

V_{IN} = 12V, V_{OUT} = 20V, I_{LOAD} = 0A to 2.5A, 150mA/µs



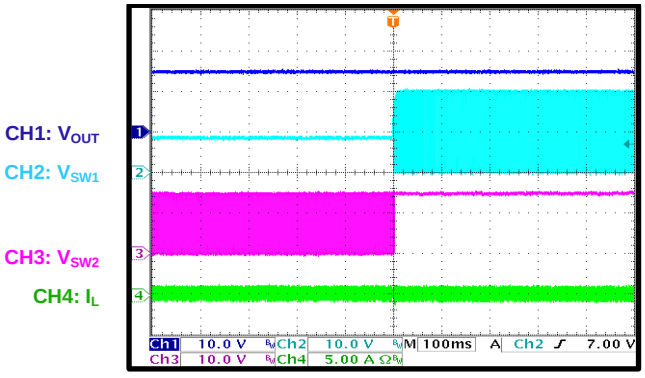
Load Transient

V_{IN} = 12V, V_{OUT} = 20V, I_{LOAD} = 2.5A to 5A, 150mA/µs



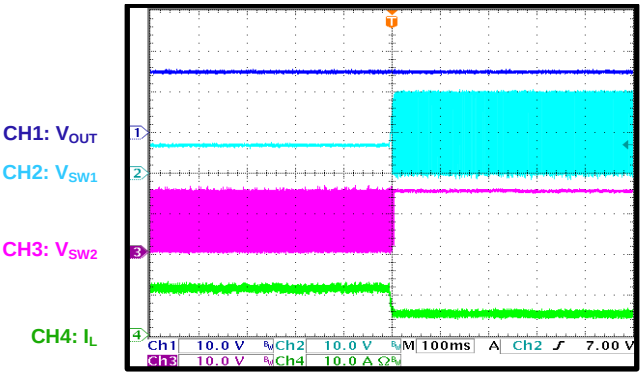
V_{IN} Transient

V_{IN} = 9V to 20V, V_{OUT} = 15V, I_{LOAD} = 0A



V_{IN} Transient

V_{IN} = 9V to 20V, V_{OUT} = 15V, I_{LOAD} = 5A



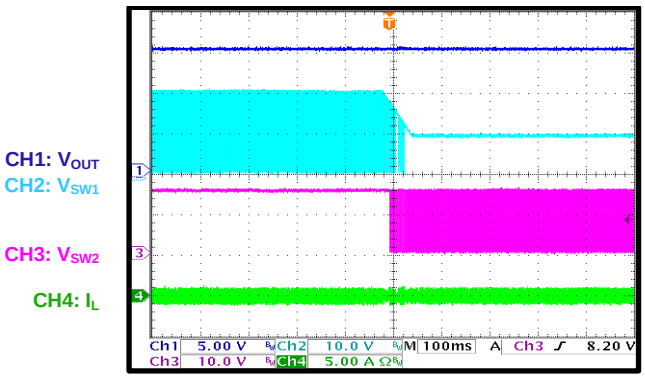
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

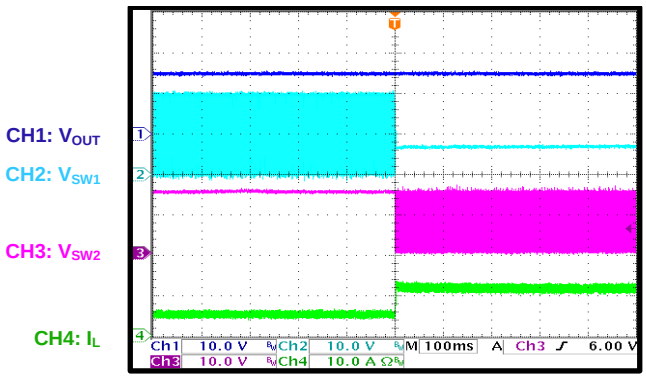
V_{IN} Transient

V_{IN} = 9V to 20V, V_{OUT} = 15V, I_{LOAD} = 0A



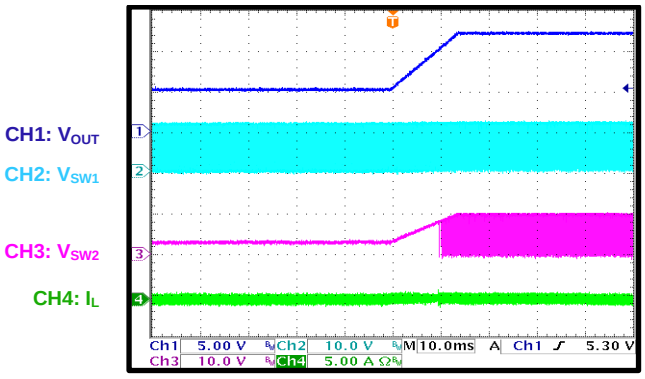
V_{IN} Transient

V_{IN} = 20V to 9V, V_{OUT} = 15V, I_{LOAD} = 5A



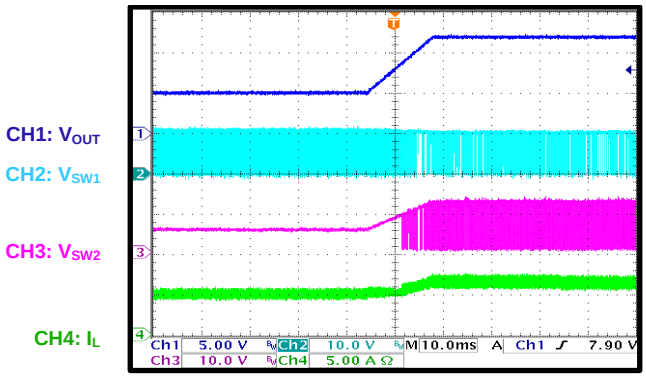
V_{OUT} Transient

V_{OUT} = 5V to 12V, I_{OUT}=0A



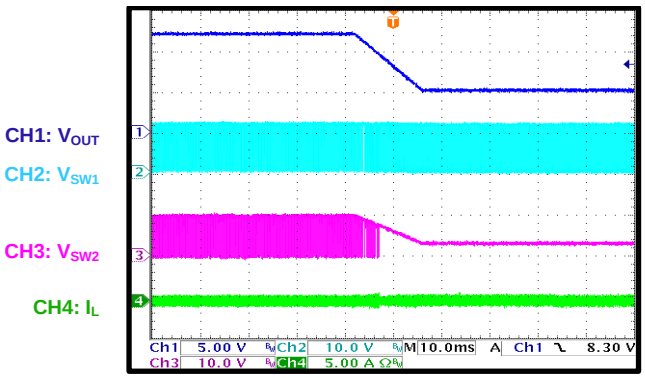
V_{OUT} Transient

V_{OUT} = 5V to 12V, I_{OUT}=5A



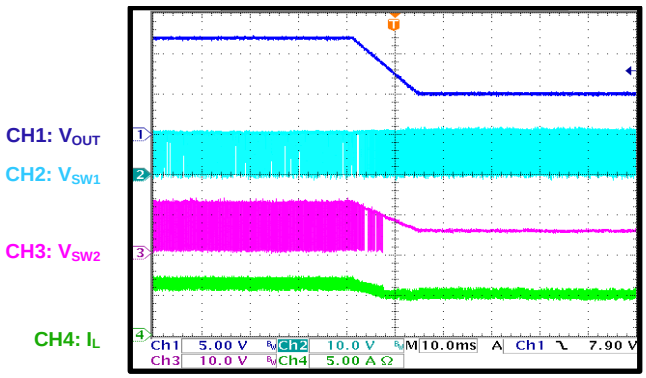
V_{OUT} Transient

V_{OUT} = 12V to 5V, I_{LOAD} = 0A



V_{OUT} Transient

V_{OUT} = 12V to 5V, I_{LOAD} = 5A



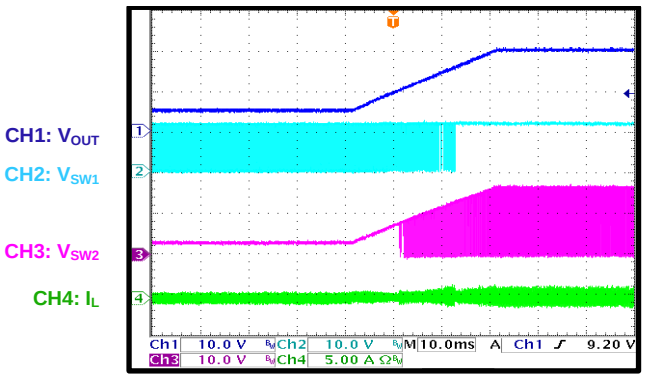
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

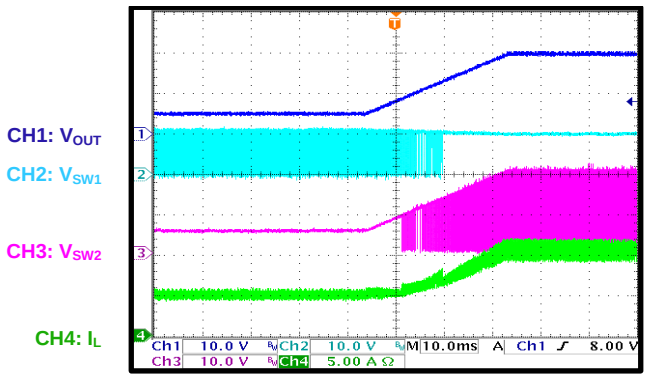
V_{OUT} Transient

V_{OUT} = 5V to 20V, I_{LOAD} = 0A



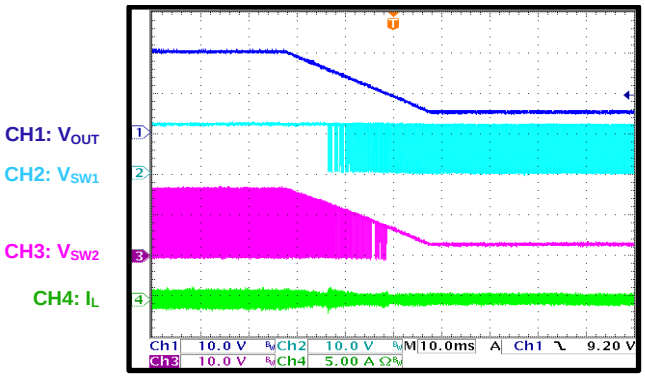
V_{OUT} Transient

V_{OUT} = 5V to 20V, I_{LOAD} = 5A



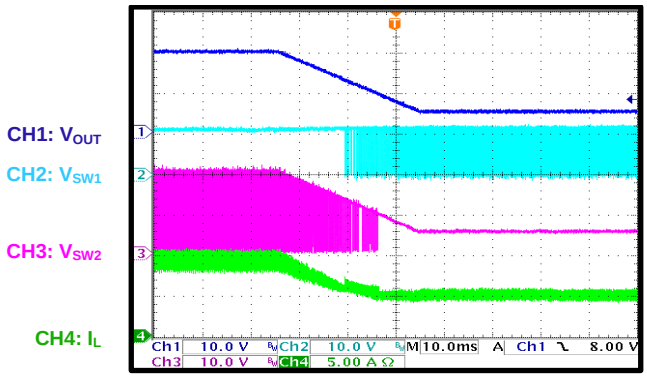
V_{OUT} Transient

V_{OUT} = 20V to 5V, I_{LOAD} = 5A

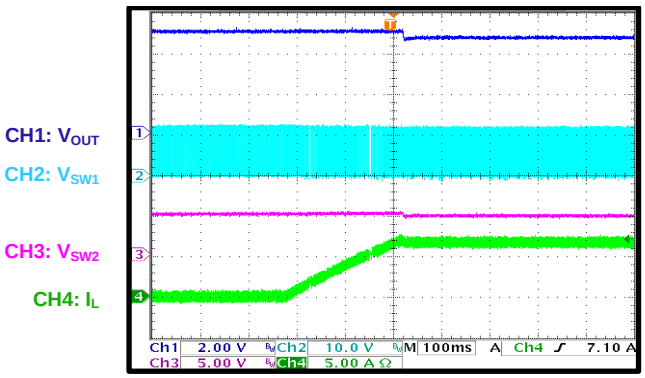


V_{OUT} Transient

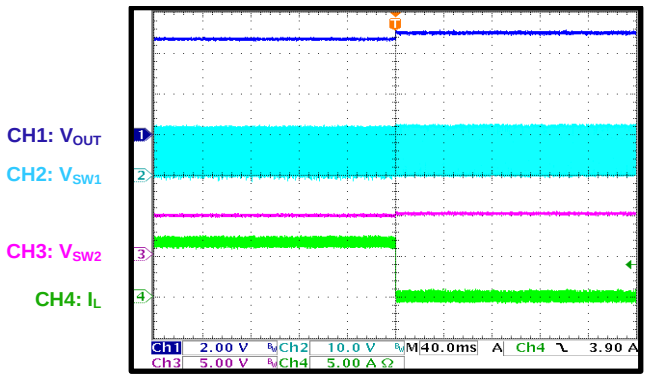
V_{OUT} = 20V to 5V, I_{LOAD} = 5A



OCP Entry



OCP Recovery

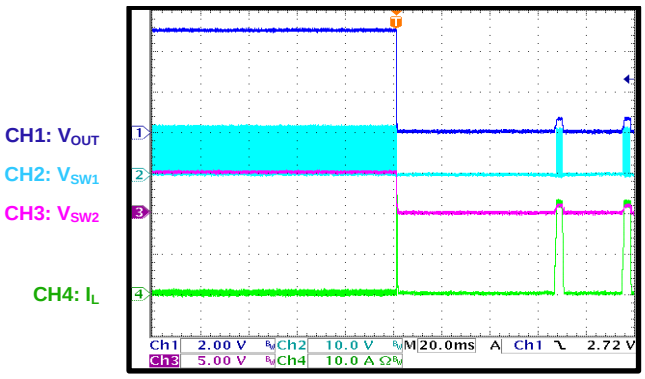


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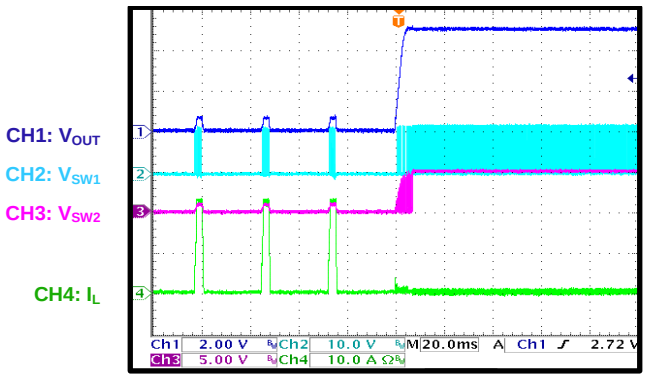
TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 12V, V_{OUT} = 5V, L = 4.7µH, T_A = 25°C, unless otherwise noted.

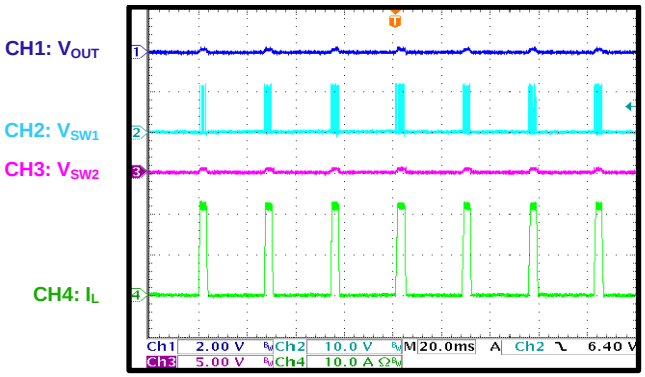
SCP Entry



SCP Recovery



SCP Steady State



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FUNCTIONAL BLOCK DIAGRAM

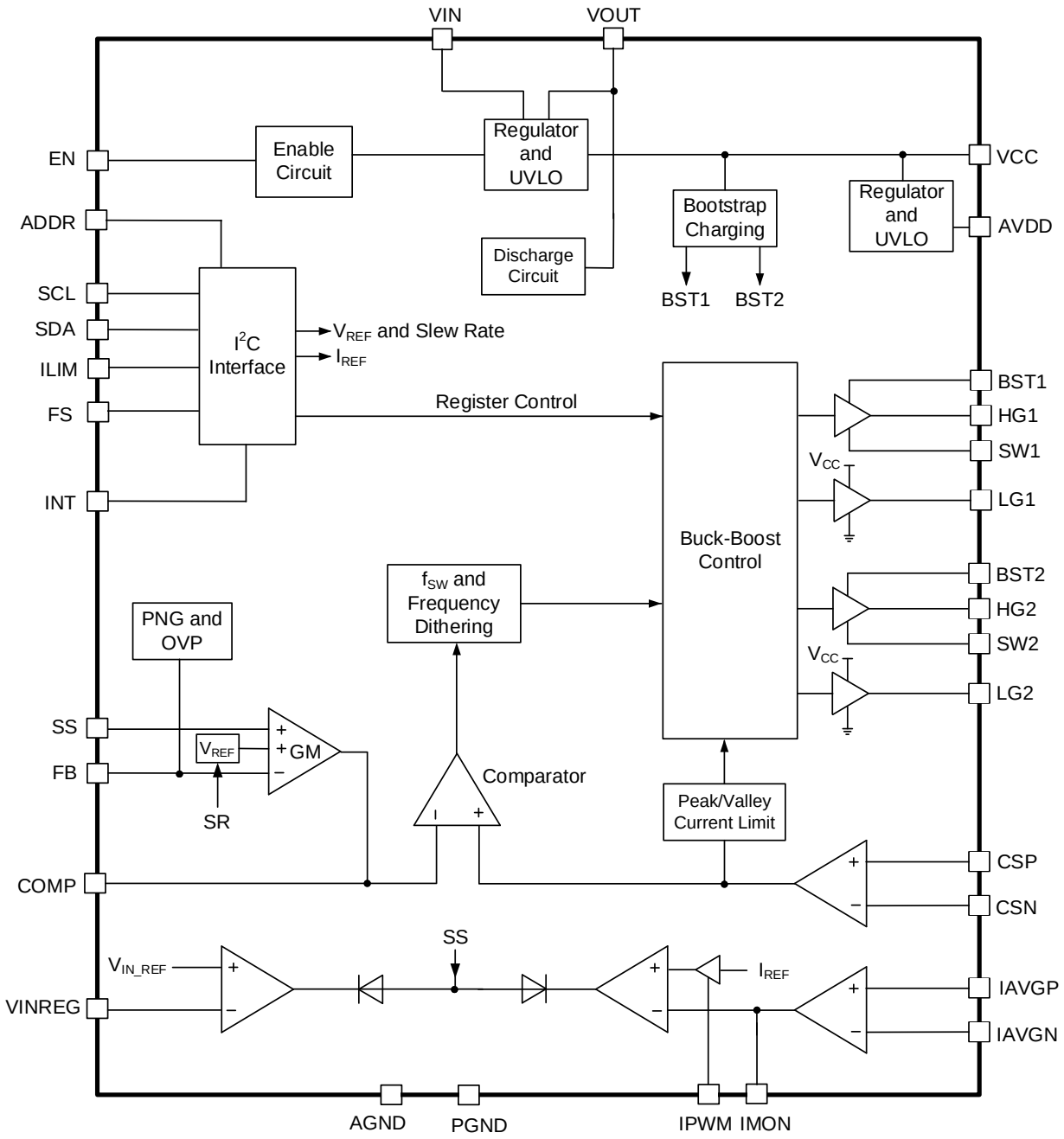


Figure 2: Functional Block Diagram

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OPERATION

The MP2984 is a high-efficiency, quad-switch, synchronous buck-boost controller. The device operates with fixed frequency in buck mode, boost mode, and buck-boost mode. Buck-boost control provides high efficiency across the entire input voltage (V_{IN}) range and smooth transient between modes.

Buck-Boost Operation

The MP2984 can regulate outputs above, below, or equal to V_{IN}. Figure 3 shows a one-inductor, four-switch power structure buck-boost topology.

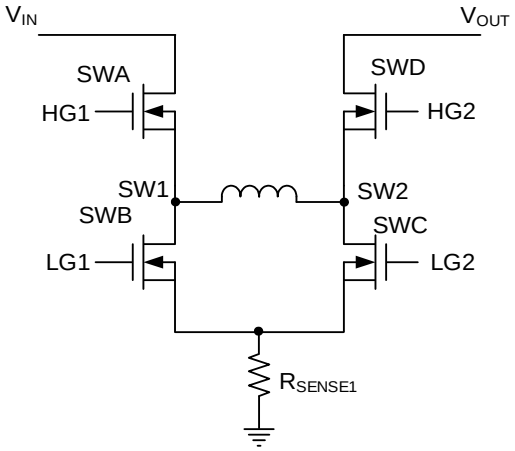


Figure 3: Buck-Boost Topology

In a buck-boost topology, the device can operate in buck mode, boost mode, or buck-boost mode with different V_{IN} inputs (see Figure 4).

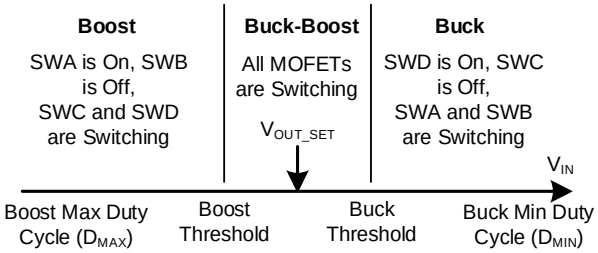


Figure 4: Buck-Boost Operating Range

Buck Mode (V_{IN} > V_{OUT})

If V_{IN} exceeds the output voltage (V_{OUT}), the device operates in buck mode. The SWA and SWB MOSFETs are switching during buck regulation. The SWC MOSFET is off, and the SWD MOSFET remains on to conduct the inductor current (I_L).

During each buck mode cycle, SWA turns on once the feedback (FB) voltage (V_{FB}) drops below the reference voltage (V_{REF}). Once SWA turns off, SWB turns on to conduct I_L until I_L rises and triggers the COMP control signal. The converter regulates V_{OUT} by repeating this operation.

Boost Mode (V_{IN} < V_{OUT})

If V_{IN} drops below V_{OUT}, the device operates in boost mode. In boost mode, SWC and SWD switch during boost regulation. Once SWB is off, SWA remains on to conduct I_L.

In each boost mode cycle, SWC turns on to conduct I_L until I_L rises and triggers the COMP control signal. SWC turns off and SWD turns on during the current freewheel. Then SWC turns on and off repeatedly to regulate V_{OUT} in boost mode.

Buck-Boost Mode (V_{IN} ≈ V_{OUT})

If V_{IN} ≈ V_{OUT}, the converter cannot provide enough power to the load in buck mode due to SWA's minimum off time (t_{OFF_MIN}). Additionally, the converter can supply too much power to load in boost mode due to SWC's minimum on time (t_{ON_MIN}). Under these conditions, the MP2984 operates in buck-boost mode to regulate V_{OUT}. Figure 5 shows the buck-boost mode waveform.

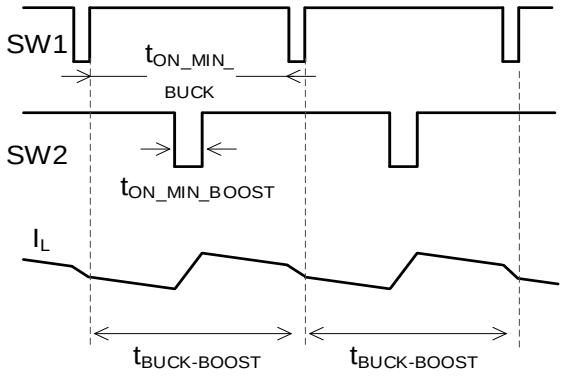


Figure 5: Buck-Boost Mode Waveform

If V_{IN} ≈ V_{OUT}, then the device operates in buck-boost mode, and a boost switch is inserted during each buck switching period. The MOSFET start-up sequence is described below:

1. SWA and SWD start up.
2. SWA and SWC start up.
3. SWA and SWD start up.
4. SWB and SWD start up.
5. I_L reaches the COMP voltage (V_{COMP}), and can supply enough current to the output.

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Power Supply

The MP2984's internal circuit is powered by AVDD (5.2V). The gate drivers are powered by VCC (7.2V). V_{IN} and V_{OUT} regulate the VCC voltage (V_{CC}). VCC powers the AVDD voltage (V_{AVDD}).

If V_{IN} is supplied and V_{EN} is high, the MP2984 regulates V_{CC} at 7.2V and V_{AVDD} to 5.2V. If V_{AVDD} exceeds the UVLO threshold and ENPWR is high, then the part starts switching and regulates V_{OUT} via soft-start control. If V_{IN} and V_{OUT} both exceed 8.8V, VCC is powered by the lower of the two voltage sources to reduce power loss. Otherwise, VCC is powered by the higher voltage power source to ensure that there is enough V_{CC}. The VCC and BST pins have separate UVLO protections that keeps the gate signal off. Ensure that VCC and BST have enough voltage to restart the part after UVLO is triggered. This does not apply to AVDD UVLO.

The MP2984 operates from a wide 6V to 36V V_{IN} range. If VCC is powered by V_{OUT} after start-up, the part operates until V_{IN} drops below 5V.

If the device is shut down through AVDD_UVLO or the EN signal, then the I²C interface cannot respond to the host and COMP is pulled low. The V_{CC}, V_{AVDD}, and the bootstrap (BST) voltage (V_{BST}) drop slowly with leakage, and all logic is off.

Start-Up

The MP2984 initiates soft start (SS) to start-up the IC. The soft-start circuit charges the SS pin, and the soft-start voltage (V_{SS}) ramps up slowly from 0V. The soft-start current (I_{SS}) is then fed to the error amplifier (EA) to control V_{OU}. After V_{SS} exceeds V_{REF} (set by the VREF bits), SS is complete, and the part enters closed-loop regulation. In steady state, V_{SS} should rise and clamp at 0.6V > V_{REF}, unless a protection has been triggered.

After start-up, the MP2984 typically enables buck switching since V_{OUT} is lower than V_{IN}. If there is a V_{OUT} voltage bias, the part should not switch until V_{SS} exceeds V_{FB}. V_{FB} is proportional to the V_{OUT} voltage bias. During SS, the IC operates in automatic pulse-width modulation (PFM) mode. Over-voltage protection (OVP) and over-current protection (OCP) with hiccup mode are not functional during SS.

Enable (EN) and Configurable Under-Voltage Lockout (UVLO) Protection

The EN pin enables and disables the MP2984. If the EN voltage (V_{EN}) exceeds its high threshold (>1.1V), then some of the MP2984's internal circuitry starts up. This is called micro-power mode. If V_{EN} exceeds its start-up threshold (1.35V), then the IC starts up and all functions are enabled. If V_{EN} drops below its low threshold (<1.28V), switching is disabled.

If V_{EN} drops below 0.4V, the MP2984 shuts down. After shutdown, the part sinks a small current from the input power (typically <1μA). EN is compatible with voltages up to 40V. For automatic start-up, connect the EN and VIN pins. During EN shutdown, the I²C interface resets to its default value after a 200ms discharge time ($t_{DISCHARGE}$).

The MP2984 features a configurable UVLO hysteresis. During start-up, the EN pin sources a 4.7μA current (see Figure 6) once V_{EN} exceeds 1.35V. V_{IN} has to decrease in order to overcome the current source and to stop switching after the IC starts up. The V_{IN} switching turn-on threshold (see Figure 6) can be calculated with Equation (1):

$$V_{IN_ON}(V)=V_{EN_ON}(V)\times(1+\frac{R_{TOP}}{R_{BOT}})=5.95V \quad (1)$$

The V_{IN} switching turn-off threshold be calculated with Equation (2):

$$V_{IN_OFF}(V)=V_{EN_OFF}(V)\times(1+\frac{R_{TOP}}{R_{BOT}})-4.7\mu A \times R_{TOP}(k\Omega)\div 1000=5.16V \quad (2)$$

Where V_{EN_ON} is about 1.35V (typically), and V_{EN_OFF} is about 1.28V.

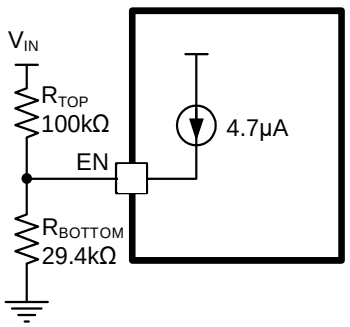


Figure 6: Configurable V_{IN} UVLO

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Forced Continuous Conduction Mode (FCCM)

The MP2984 can operate in forced continuous conduction mode (FCCM). The buck on time (t_{ON_BUCK}) and boost off time (t_{OFF_BOOST}) are determined by an internal circuit. This is to have the part operate with a fixed switching frequency (f_{SW}), based on the V_{IN} / V_{OUT} ratio. If the load decreases, then the input current (I_{LIMIT}) drops, which may cause I_L to go negative from V_{OUT} to V_{IN} during SWD's on time (t_{ON}). This forces I_L to operate with a fixed f_{SW} in FCCM, which produces a low V_{OUT} ripple.

Switching Current Limit

The MP2984 senses the LS-FET current through a current-sense resistor (R_{SENSE1}) in the loop control. This allows the device to provide valley current limiting in buck mode and peak current limiting in boost mode, during each cycle-by-cycle switching period. In buck mode, the next period does not start until I_L drops to the valley current limit, so the device may fold back the frequency when the valley current limit is triggered. The buck current limit is fixed at 133mV and the boost current limit is fixed at 150mV. For example, if R_{SENSE1} = 5mΩ, the buck current limit is 133mV/5mΩ = 26.6A, and the boost current limit is 150mV/5mΩ = 30A.

The switching current limit can be configured by the external sense resistor. The SWB and SWC current signal is internally blanked for about 180ns to enhance noise rejection.

Under over-current (OC) conditions, the MP2984 runs with a cycle-by-cycle current limit. The device may initiate hiccup protection or a latch-off protection based on OCP_MODE.

In hiccup mode, the device turns off once V_{FB} drops below 60% of V_{REF}, and triggers the switching current limit after the SS period. The device recovers after a fixed off time, which is configured by the SS capacitor discharge period.

In latch-off mode, the MP2984 turns off if V_{FB} drops below 60% of V_{REF}, and triggers the switching current limit after the SS period. The device recovers after the next power cycle.

If hiccup and latch-off protection are disabled, the device continues switching with the cycle-by-cycle current limit. Hiccup and latch-off protections are masked during soft start.

Average Current Limit

The IAVGP and IAVGN pins sense the output current. A current-sense resistor (R_{SENSE2}) can be connected to the VOUT line to control the average output current limit. Once the sensed signal exceeds the current limit reference voltage, one internal EA pulls down V_{SS}. Eventually, V_{SS} replaces V_{REF} to control COMP instead of V_{REF}, and the inductor current is limited by COMP to transfer less energy to output. The SS pin keeps V_{OUT} low until the average load current drops.

If the switching current is regulated by the average current limit, and it does not trigger a cycle-by-cycle current limit, the MP2984 does not run hiccup or latch-off protection, even if the average current limit is reached. This allows the device to maintain a constant current charge.

The ILIM bits provide 8 selectable current limits, and the external IPWM pin provides high-resolution current limit adjusting.

If the IPWM pin is high or floating, the average output current limit is determined by both ILIM bits setting and R_{SENSE2}.

Another method to set the average output current limit is by multiplying the value set by the ILIM bit by the IPWM duty cycle. When the ILIM bits are set to 111 and the current limit is configured by the IPWM pin, the current limit threshold (OCP_LIMIT) can be calculated with Equation (3):

$$OCP_LIMIT\ (mV) = 65.8mV \times DUTY + 4.49mV \quad (3)$$

Where OCP_LIMIT is the average load current limit configured by the IPWM pin (when the ILIM bits = 111), and DUTY is IPWM pin input signal duty cycle, which can be between 0 and 0.9.

The real output current limit (in A) is determined by OCP_LIMIT / R_{SENSE2}.

For example, if the IPWM signal duty cycle is 48%, the final average output current limit is about 36mV. The IPWM signal frequency can be between 5kHz and 100kHz. It is recommended to have a 20kHz frequency signal. For USB PD applications, it is recommended for R_{SENSE2} to be 10mΩ.

Overload Protection (OLP) and Short-Circuit Protection (SCP)

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If an overload (OL) fault occurs, the MP2984 limits I_{OUT} via I_{LIMIT} loop regulation. If the I_{LIMIT} is disabled, then I_{OUT} is limited cycle-by-cycle. If the IC is operating in boost mode during cycle-by-cycle current limiting, then the SWC peak current (I_{PEAK}) is limited. If the IC is operating in buck mode, then SWB remains on until I_L reaches I_{LIMIT_VALLEY}. The next cycle begins once I_L reaches I_{LIMIT_VALLEY}. Therefore, I_L can be controlled in all operation modes.

Output Voltage Regulation

The MP2984 regulates V_{OUT} via the FB pin. V_{FB} is compared to the internal V_{REF}. V_{REF} can be between 300mV to 2.047V, depending on the VREF register bit settings. The EA output on COMP controls I_L to V_{OUT}.

Switching Frequency (f_{SW}) and Frequency Spread Spectrum (FSS)

The MP2984 configures the switching frequency (f_{SW}) via the 2-bit FSW register. F_{SW} can be set to 200kHz, 300kHz, 400kHz, or 600kHz via the 2-bit FSW register. For most applications, it is recommended to set f_{SW} to 400kHz.

The MP2984 features frequency spread spectrum (FSS) and frequency dithering. Set the Dither bit to 1 (0x02h, bit D[4]) to enable FSS and frequency dithering. Set the Dither bit = 0 to disable FSS and frequency dithering. FSS reduces the peak emissions at certain frequencies.

The MP2984 uses a 2kHz triangle wave to modulate the internal oscillator. The frequency span of the spread spectrum is ±6%.

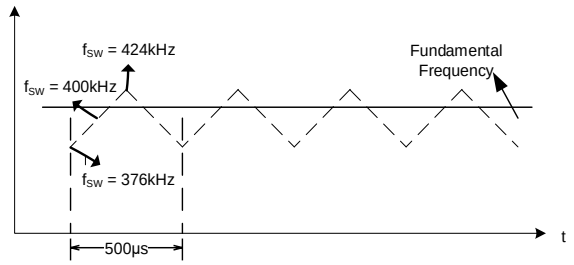


Figure 7: Frequency Spread Spectrum

FSS can operate with a 200kHz, 300kHz, 400kHz, or 600kHz f_{SW}.

Gate Driver and Bootstrap (BST) Power

The MP2984 provides four N-channel power MOSFET gate drivers for the H-bridge MOSFETs (see Figure 3 on page 24). Each driver is capable of sourcing and sinking currents. In buck mode, LG1 and HG1 are switching while HG2 remains on. In boost mode, LG2 and HG2 are switching while HG1 remains on. LG1 and LG2 are powered by V_{CC}, while HG1 and HG2 are powered by V_{BST1} and V_{BST2}.

Capacitors between BST1 and SW1, as well as BST2 to SW2 are necessary to supply power, which can be supplied via an internal diode from VCC or from charging each other.

Over-Voltage Protection (OVP)

The MP2984 monitors the feedback voltage (V_{FB}). If V_{FB} exceeds 127% of V_{REF} and the OVP_MODE bits are set to 01, then the IC discharges the output capacitor (C_{OUT}) via an internal discharge resistor. Discharging stops once V_{FB} drops to 111% of the regulation voltage. If the OVP_MODE bits are set to 00, then there is no logic available to stop switching, even if V_{FB} is above the OVP threshold. If the OVP_MODE bits are set to 10, the IC latches off once V_{OUT} reaches 127% of V_{REF}.

Interrupt (INT)

The interrupt (INT) pin is an open-drain output that indicate an interrupt signal for the following fault conditions : over-current fault (OC), over-voltage (OV) fault, over-temperature (OT) fault, and V_{OUT} is not within its normal range.

If OCP, output OVP, or thermal shutdown is triggered, the corresponding register bit (OCP, bit D[1], OVP, bit D[2], or OTP, bit D[4]) is set to 1. INT pulls low to indicate an interrupt signal, depending on the related mask register setting. If the device is off, then INT is an open drain.

Current Monitor Output (IMON)

The MP2984 senses the average I_{LOAD} via a current-sense resistor, and outputs a voltage signal on the IMON pin. The signal is amplified by the voltage difference between the IAVGP and IAVGN pins (V_{IAVGP} - V_{IAVGN}). Connect a small capacitor connected between the IMON and AGND pins. The IMON voltage (V_{IMON}) can be calculated with Equation (4):

$$V_{IMON}\ (mV) = Gain \times I_{OUT}\ (A) \times R_{SENSE2}\ (m\Omega)\ (4)$$

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Where Gain is typically 18.8mV/V, and R_{SENSE2} is the average load's current-sense resistor.

Configurable Soft-Start Time (t_{SS})

The SS pin sets the soft-start time (t_{SS}). t_{SS} can be estimated with Equation (5):

$$t_{SS}\,(ms) = C_{SS}\,(nF) \times V_{REF}\,(V) / I_{SS}\,(\mu A) \quad (5)$$

Where the I_{SS} charge current is about 6μA, the soft-start capacitor (C_{SS}) is 47nF, and V_{REF} is 0.5V. This means that t_{SS} is about 3.9ms.

Slew Rate Control and Output Discharge

The internal SR bits set the V_{OUT} slew rate. Four The V_{REF} slew rate (rising and falling) can be set to 38mV/ms, 50mV/ms, 75mV/ms, or 150mV/ms.

During voltage transient, the discharge path is function once GO_BIT is set to 1. The discharge path is disabled once GO_BIT resets to 0 (e.g. after the V_{REF} change is complete). If V_{OUT} is not discharged to the target voltage before the V_{REF} change is complete (e.g. due to a large C_{OUT}), then the OVP discharge function or the DISCHG bit can be used to continue discharging V_{OUT}.

V_{OUT} can be discharged via the following:

- GO_BIT = 1. The discharge path is enabled, then GO_BIT resets to 0 after a 20ms delay.
- The DISCHG bit = 1.
- OVP_MODE bits = 01, and V_{FB} exceeds 127% of V_{REF}.
- The ENPWR bit is off. The discharge path is enabled for 200ms.
- The EN pin is off. The discharge path is enabled for 200ms.
- If V_{IN} UVLO protection is triggered and some residual V_{AVDD} remains, then the discharge path is enabled for 200ms. This discharge path may be disabled if the V_{AVDD} drops.

Current Limit Control (ILIM)

During start-up, the ILIM pin status is latched to the register status, and the ILIM pin sets the default ILIM bit values. After start-up, the pin status change should not affect the register status, unless the pin status is changed or the part shuts down.

Frequency Control (FS)

During start-up, the FS pin status is latched to the register status, and the FS pin sets the default FSW bit values. After start-up, the pin status change should not affect the register status, unless the pin status is changed or the part shuts down.

V_{IN} Regulation Loop (VINREG)

The VINREG pin sets the minimum operating V_{IN}. If V_{IN} drops to the set voltage level, then SS is pulled down to decrease I_{OUT} and maintain V_{IN}. If V_{IN} drops below the VINREG threshold, the IC stops converting power to V_{OUT}.

Thermal Shutdown

Thermal shutdown prevents the device from operating at exceedingly high temperatures. If the junction temperature (T_J) exceeds 150°C, then the MP2984 shuts down. Once T_J drops below 125°C, then the IC initiates a SS to resume normal operation. If thermal shutdown is triggered and the INT pin is not masked, the INT pin is pulled low.

I²C Interface

The I²C interface has four device addresses (defined as 1100xxx) that are set via a resistor divider between the ADDR and AVDD pins. This address is 7 bits long, and is followed by an 8th read/write (R/W) data direction bit. A 1 indicates a read (R) command, and a 0 indicates a write (W) command (R = 1 and W = 0). The 7-bit address acts as a slave address, and can support both standard mode (100kbps) communication and fast mode (400kbps) communication.

Table 1 shows the configurable device address settings. The ADDR pin also affects the ENPWR bit default value.

Table 1: Device Address Setting

Device Address	R_{TOP}	R_{BOT}	Default ENPWR
60h (1100 000)	-	0Ω	1
62h (1100 010)	100kΩ	59kΩ	1
64h (1100 100)	68kΩ	100kΩ	0
66h (1100 110)	0Ω	-	0

For more details, see I²C Registers section on page 31.

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I²C Data Transfer

Every byte on the SDA line should be 8 bits long. Each byte should be followed by an acknowledge (ACK) bit. The ACK clock pulse is generated by the master address. The transmitter releases the SDA line (SDA = high) during the ACK clock pulse. The SDA line is pulled down by the receiver during the ACK clock pulse to stabilize the SDA line (SDA = low) during the high period of this clock pulse.

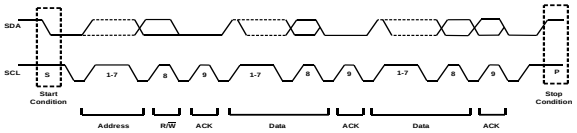


Figure 8: Complete Data Transfer

Figure 8 shows the complete data transfer format. After a start (S) condition, a slave address is sent. This address is 7 bits long, and is followed by an 8th data direction bit (R/W). A 0 indicates a transmission (W), and a 1 indicates a request for data (R). A data transfer is always terminated by a stop (P) condition, which is generated by the master. If a master needs to communicate on the bus, then the master can generate a repeated start (S) condition and address another slave without having to generate a stop (P) condition.

The MP2984 includes a full I²C slave controller. The I²C slave complies with the I²C specification requirements. For a single data updated, the I²C slave requires a start (S) condition, valid I²C address, register address byte, and data byte. After receiving each byte, the MP2984 acknowledges by pulling the SDA line low during the high period of a single clock pulse. A valid I²C address selects the MP2984. The MP2984 then performs an update on the falling edge of the LSB byte.

Figure 9 and Figure 10 show examples of the I²C read and write commands.

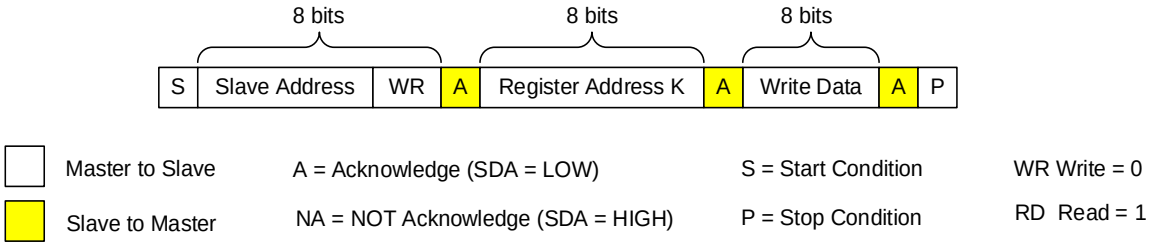


Figure 9: I²C Write Example

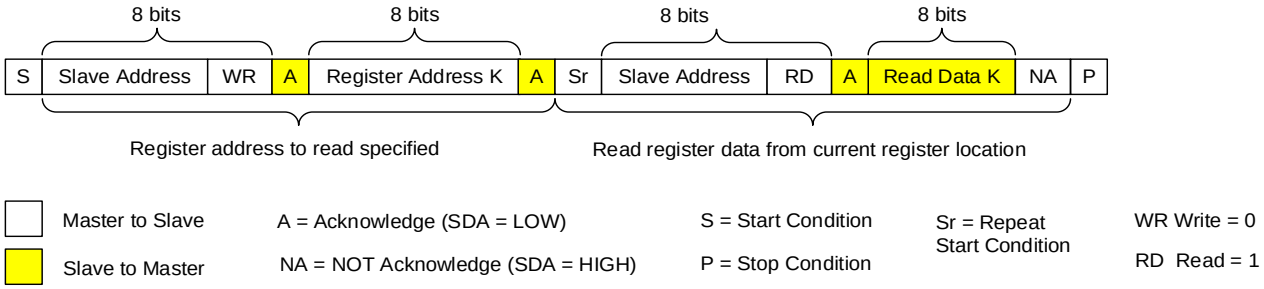


Figure 10: I²C Read Example

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REGISTER DESCRIPTION

Register Map

Address	Register	Type	D7	D6	D5	D4	D3	D2	D1	D0	Reset State
0x00	REF_LSB	R/W	N/A	N/A	N/A	N/A	N/A	VREF_L			0000 0100
0x01	REF_MSB	R/W	VREF_H								0011 1110
0x02	CONTROL_1	R/W	SR		DISCHG	DITHER	PNG_ LATCH	RESERVED (10)	GO_BIT	ENPWR	0100 010x (11)
0x03	CONTROL_2	R/W	FSW		N/A	BB_FSW	OCP_MODE		OVP_MODE		xx00 0101 (11)
0x04	ILIM	R/W	N/A	N/A	N/A	N/A	N/A	ILIM			0000 1xxx (11)
0x05	INTERRUPT_ STATUS	R/W	N/A	N/A	N/A	OTP	CC	OVP	OCP	PNG	0000 0000
0x06	INTERRUPT_ MASK	R/W	N/A	N/A	N/A	M_OTP	M_CC	M_OVP	M_OCP	M_PNG	0000 0001

Notes:

10) Reserved bits should not be set to a different value in application.
11) "x" means the value is determined by the external pin set during start-up. Refer to below register function description for details.

Feedback Reference Data Format (12)

Name	VREF														
Format	Direct, unsigned binary integer														
Register Name	N/A					VREF_H, bits D[7:0]							VREF_L, bits D[2:0]		
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	N/A					R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Function	N/A					Data bit is high								Data bit is low		
Default (0.5V)	N/A					500 integer										

Note:

12) There are 11 bits that can set V_{REF}. If V is an 11-bit, unsigned binary integer of VREF, bits[10:0], then V_{FB} = V / 1000.

I²C REGISTERS

Register 0x00: REF_LSB (Read/Write)

Bits	Name	Default	Description
D[2:0]	VREF_L	100	Feedback (FB) reference voltage (V_{REF}) lower 3 bits. LSB = 1mV.

Register 0x01: REF_MSB (Read/Write)

Bits	Name	Default	Description
D[7:0]	VREF_H	0011, 1110	FB V_{REF} higher 8 bits. LSB = 8mV.

Register 0x02: CONTROL (Read/Write)

Bits	Name	Default	Description
D[7:6]	SR	01	Sets the V_{OUT} slew rate. The SR control is only functional once soft start (SS) is complete. During the SS time (t_{SS}), the output voltage (V_{OUT}) slew rate is controlled by SS. V_{OUT} slew rate = V_{REF} slew rate x feedback ratio. Where the feedback ratio is (R1 + R2) / R2. 00: 38mV/ms slew rate 01: 50mV/ms slew rate 10: 75mV/ms slew rate 11: 150mV/ms slew rate

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Bits	Name	Default	Description
D[5]	DISCHG	0	Enables the output discharge path. The DISCHG bit is functional even if the ENPWR bit is low. 0: If this bit is set to 0, then the discharge path is disabled 1: If this bit is set to 1, then V_{OUT} is discharged via an internal resistor This bit does not affect the output discharge function for the following conditions: V_{OUT} is set via the I²C, ENPWR is off, the EN pin is off, output over-voltage protection (OVP) is triggered (OVP_MODE enables the discharge path), and V_{IN} under-voltage lockout (UVLO) protection is triggered. If GO_BIT = 1, then the discharge path is enabled. After a 20ms delay, GO_BIT resets to 0 and the discharge path is disabled. For most applications, it is recommended to set a low slew rate so that V_{OUT} can follow V_{REF} via the internal discharge current. If V_{OUT} cannot follow V_{REF} due to a large output capacitor (C_{OUT}), then an additional 20ms of discharge time is added.
D[4]	DITHER	0	Enables frequency dithering. 0: If this bit is set to 0, then frequency dithering is disabled 1: If this bit is set to 1, then frequency dithering is enabled
D[3]	PNG_ LATCH	0	Resets the PNG status bit. For more details, see the PNG bit description on page 34. 0: PNG bit status recovers to 0 once V_{OUT} is within to its normal voltage range 1: PNG bit status latches at 1 once V_{OUT} exceeds the power good voltage (V_{PG}) range
D[2]	RESERVED	1	Reserved. Do not write a different value to this bit in application.
D[1]	GO_BIT	0	Enables the V_{REF} output change function. Write the VREF_L and VREF_L registers, and then write GO_BIT to 1. V_{REF} and V_{OUT} are determined by the new V_{REF} value. The host reads GO_BIT to determine whether V_{REF} scaling is complete. The output discharge path is enabled once GO_BIT = 1, regardless of what the DISCHRG bit's setting. Once GO_BIT resets to 0, output discharge continues. The discharge path is disabled after a 20ms delay. 0: V_{OUT} cannot be adjusted 1: V_{OUT} is determined by the VREF registers. Once V_{REF} the level set by the VREF bits, then GO_BIT resets to 0. This setting prevents false V_{OUT} scaling
D[0]	ENPWR	x	Enables the MP2984's power MOSFETs. ENPWR's default value is determined by the ADDR pin setting. For more details, see Table 1 on page 28. If ENPWR is reset to 0, then the VREF bits are reset to 0011 1110 100 (0.5V). The VREF bits should only reset one time, so that once ENPWR = 0, V_{REF} can adjust to the value set by the I²C. After ENPWR = 0, output discharge is functional for 200ms. 0: Power MOSFETs are disabled 1: Power MOSFETs are enabled

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I²C V_{REF} Sequence for USB Power Delivery (PD) Applications

If the sink device is unplugged, then the PD controller sets ENPWR to 0 in order to disable the bus voltage (V_{BUS}). Once the sink device is attached again, the PD controller uses the following sequence to set V_{BUS} to 5V:

1. Write the VREF bits based on the new V_{OUT}.
2. Write GO_BIT = 1, and keep ENPWR = 0. This can set V_{REF} to the target value.
3. To enable V_{OUT}, write ENPWR = 1.

Figure 11 shows the request sequence for a new V_{OUT}.

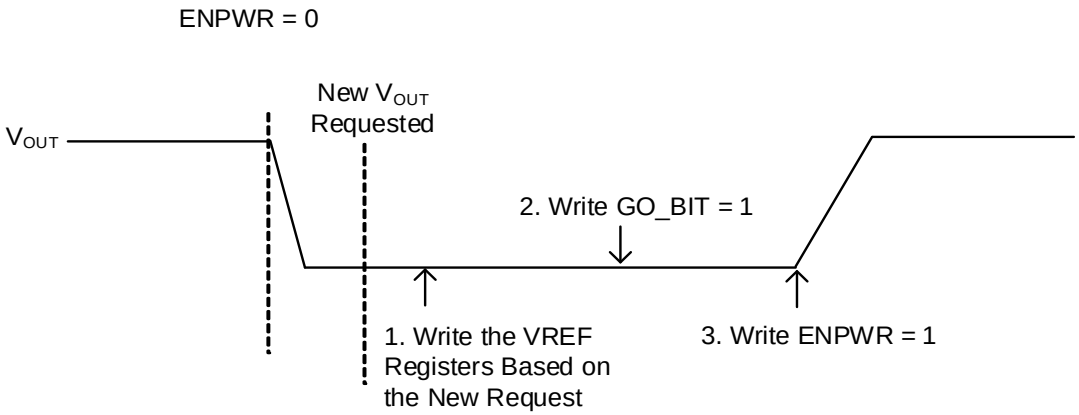


Figure 11: New V_{OUT} Request Sequence

Register 0x03: CONTROL_2 (Read/Write)

Bits	Name	Default	Description
D[7:6]	FSW	00, 10	Sets the switching frequency (f_{SW}) slew rate. FSW's default value is determined by the FS pin setting during start-up. For different voltage level settings, see the Electrical Characteristics section on page 9. 00: 200kHz slew rate 01: 300kHz slew rate 10: 400kHz slew rate 11: 600kHz slew rate
D[4]	BB_FSW	0	Sets the buck-boost f_{SW}. 0: Sets the buck-boost f_{SW} to 40% of the base f_{SW} 1: Sets buck-boost f_{SW} to 80% of the base f_{SW}
D[3:2]	OCP_ MODE	01	Sets the mode once the current reaches the cycle-by-cycle valley current limit (I_{LIMIT_VALLEY}) in buck mode or the cycle-by-cycle peak current limit (I_{LIMIT_PEAK}) in boost mode. 00: No hiccup mode or latch-off protection; inductor current (I_L) is limited cycle-by-cycle 01: The part enters hiccup mode once the current reaches I_{LIMIT}; V_{FB} < 60% of V_{REF}, t_{SS} is controlled by the SS discharge path 10: OCP is triggered; the part initiates a SS to resume normal operation 11: Reserved

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Bits	Name	Default	Description
D[1:0]	OVP_ MODE	01	Sets the mode once the voltage exceeds the OVP threshold (127% of V_{REF}). 00: No protection or discharge after OVP; V_{OUT} is regulated by COMP 01: V_{OUT} is discharged via an internal resistor. The IC stops switching once V_{FB} exceeds 127% of V_{REF}, and recovers once V_{FB} drops below 111% of V_{REF} 10: Over-voltage (OV) fault has occurred, and OVP is triggered (latch-off protection); no discharge after OVP 11: Reserved

Register 0x04: ILIM (Read/Write)

Bits	Name	Default	Description
D[2:0]	ILIM	N/A	Average I_{LIMIT}. The average I_{LIMIT} can configure the output current limit (I_{OUT_LIMIT}). ILIM's default value is determined by the ILIM pin setting during start-up. Float the ILIM pin to set the ILIM bits to 111. For different voltage level settings, see the Electrical Characteristics section on page 9. 000: 26mV I_{LIMIT}, 2.6A I_{LIMIT} with 10mΩ R_{SENSE2} 001: 32mV I_{LIMIT}, 3.2A I_{LIMIT} with 10mΩ R_{SENSE2} 010: 38mV I_{LIMIT}, 3.8A I_{LIMIT} with 10mΩ R_{SENSE2} 011: 45mV I_{LIMIT}, 4.5A I_{LIMIT} with 10mΩ R_{SENSE2} 100: 50mV I_{LIMIT}, 5A I_{LIMIT} with 10mΩ R_{SENSE2} 101: 56mV I_{LIMIT}, 5.6A I_{LIMIT} with 10mΩ R_{SENSE2} 110: 62mV I_{LIMIT}, 6.2A I_{LIMIT} with 10mΩ R_{SENSE2} 111: 68mV I_{LIMIT}, 6.8A I_{LIMIT} with 10mΩ R_{SENSE2}

Register 0x05: INTERRUPT_STATUS (Read/Write)

Bits	Name	Default	Description	Reset Conditions
D[4]	OTP	0	Indicates whether an over-temperature (OT) fault has occurred. 0: No fault has occurred 1: OT fault has occurred; the IC enters over-temperature protection (OTP)	These bit are latched once triggered. Write 0xFF to the register to reset the interrupt status and the INT pin's state.
D[3]	CC	0	Indicates the output average current limit. 0: No fault has occurred 1: The output current exceeds the average current limit reference, and V_{OUT} drops	
D[2]	OVP	0	Indicates whether an output OV fault has occurred. 0: No fault has occurred 1: OV fault has occurred; the IC enters OVP, and I_{OUT_LIMIT} exceeds the I_{LIMIT} reference so that V_{OUT} drops below the OVP falling threshold	
D[1]	OCP	0	Indicates whether over-current protection (OCP) has been triggered. 0: No fault has occurred 1: OCP is triggered; V_{FB} < 60% of V_{REF}, and SS is complete	

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Bits	Name	Default	Description	
D[0]	PNG	0	Indicate whether V_{OUT} is within its normal range (0.5V to 28V). The PNG_LATCH bit controls the PNG reset behavior. 0: V_{OUT} is within its normal range (output power is good) 1: V_{OUT} is outside of its normal range (output power is not good); the IC indicates once V_{OUT} is within 0.5V to 28V	If PNG_LATCH = 0, then this bit indicates an instantaneous value and INT indicates an instantaneous state. If PNG_LATCH = 1, then this bit is latched once triggered. Write 0xFF to the register to reset the interrupt status and the INT pin's state.

Register 0x06: INTERRUPT_MASK (Read/Write)

Bits	Name	Default	Description
D[4]	M_OTP	0	Determines when an OTP mask bit is masked or not masked. The settings listed below are similar for other mask bits. 0: Not masked 1: Masked (only the INT input is masked)
D[3]	M_CC	0	Determines when a CC mask bit is masked or not masked. 0: Not masked 1: Masked (only the INT input is masked); the CC bit cannot be masked
D[2]	M_OVP	0	Determines when an OVP mask bit is masked or not masked. 0: Not masked 1: Masked (only the INT input is masked)
D[1]	M_OCP	0	Determines when an OCP mask bit is masked or not masked. 0: Not masked 1: Masked (only the INT input is masked)
D[0]	M_PNG	1	Determines when a PNG mask bit is masked or not masked. 0: Not masked 1: Masked (only the INT input is masked)

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APPLICATION INFORMATION

Output Voltage Setting

The default V_{OUT} is set via the FB resistor divider. The default V_{REF} is 0.5V. The bottom resistor in the resistor divider is typically between 1kΩ and 50kΩ. The top resistor (R1) in the resistor divider can be calculated with Equation (6):

$$R1 = \frac{V_{OUT} - V_{REF}}{V_{REF}} \times R2 \quad (6)$$

The I²C interface can also select V_{REF} to set V_{OUT}.

Selecting the Inductor

The inductor selection is based on the operation mode (buck mode, boost mode, or buck-boost mode). The buck mode inductance (I_{L_BUCK}) can be calculated with Equation (7):

$$I_{L_BUCK} = \frac{V_{OUT}}{f_{SW} \times \Delta I_L} \times (1 - \frac{V_{OUT}}{V_{IN}}) \quad (7)$$

Where ΔI_L is the peak-to-peak inductor ripple current.

Choose ΔI_L to be about 30% to 50% of the maximum load current (I_{LOAD_MAX}).

In boost mode, the inductor should be determined by ΔI_L. Choose ΔI_L to be about 30% to 50% of the maximum input current (I_{IN_MAX}). The boost mode inductance (I_{L_BOOST}) can be calculated with Equation (8):

$$I_{L_BOOST} = \frac{V_{IN} \times (V_{OUT} - V_{IN})}{V_{OUT} \times f_{SW} \times \Delta I_L} \quad (8)$$

The maximum input current (I_{IN_MAX}) can be calculated with Equation (9):

$$I_{IN_MAX} = \frac{V_{OUT} \times I_{LOAD_MAX}}{V_{IN} \times \eta} \quad (9)$$

Where ΔI_L is about 30% to 50% of I_{IN_MAX}, and η is the efficiency.

A larger inductor reduces ΔI_L; however, a larger inductor is takes up more space and can reduce the converter's achievable bandwidth by moving the right half-plane zero to a lower frequency. Choose an inductor that matches the specific application requirements.

Selecting the Input Capacitor (C_{IN})

In buck mode, the converter has a discontinuous input current (I_{IN}), and requires a capacitor to supply AC current to the converter while maintaining the DC V_{IN}. (In boost mode, the converter has a continuous I_{IN}.) It is recommended to use ceramic capacitors placed as close to VIN as possible for the best performance. Ceramic capacitors with X5R or X7R dielectrics are recommended due to their low ESR and small temperature coefficients. Choose a capacitor with a ripple current rating greater than the maximum input ripple current (ΔI_{IN}). The buck mode input ripple current (I_{CIN_RMS}) can be estimated with Equation (10):

$$I_{CIN_RMS} = I_{OUT} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})} \quad (10)$$

The worst-case condition in buck mode occurs at V_{IN} = 2 x V_{OUT}, which can be calculated with Equation (11):

$$I_{CIN_RMS} = \frac{I_{OUT}}{2} \quad (11)$$

For simplification, choose an input capacitor (C_{IN}) with an RMS current rating greater than half of I_{LOAD_MAX}.

C_{IN} determines the converter's (ΔV_{IN}). If there is a ΔV_{IN} in the system, choose a C_{IN} that meets the system specification.

In buck mode, ΔV_{IN} can be estimated with Equation (12):

$$\Delta V_{IN} = \frac{I_{OUT}}{f_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}}) \quad (12)$$

The worst-case condition occurs at V_{IN} = 2 x V_{OUT}, which can be calculated with Equation (13):

$$\Delta V_{IN} = \frac{1}{4} \frac{I_{OUT}}{f_{SW} \times C_{IN}} \quad (13)$$

Selecting the Output Capacitor (C_{OUT})

In boost mode, the converter has a discontinuous output voltage (I_{OUT}). The output

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capacitor (C_{OUT}) should be capable of reducing the output voltage ripple (ΔV_{OUT}).

A higher C_{OUT} may be required to lower ΔV_{OUT} and transient response. Ceramic capacitors with low ESR are recommended for their small size and low output voltage ripple. For ceramic capacitors, the capacitance dominates the impedance at f_{SW} and causes the majority of ΔV_{OUT}. For simplification, (ΔV_{OUT}) can be estimated with Equation (14):

$$\Delta V_{OUT} = \frac{(1-\frac{V_{IN}}{V_{OUT}})\times I_{LOAD}}{C_{OUT}\times f_{SW}} \quad (14)$$

For hybrid, polymer, or electrolytic capacitors, the ESR dominates the impedance at f_{SW}. For simplification, ΔV_{OUT} can be estimated with Equation (15):

$$\Delta V_{OUT} = \frac{(1-\frac{V_{IN}}{V_{OUT}})\times I_{LOAD}}{C_{OUT}\times f_{SW}} + \frac{I_{LOAD}\times R_{ESR}\times V_{OUT}}{V_{IN}} \quad (15)$$

For 100W USB PD applications, it is recommended to use a 100µF polymer capacitor and four 22µF ceramic capacitors.

Choose a C_{OUT} that satisfies the ΔV_{OUT} and load transient requirements of the design. Capacitance derating should be taken into consideration when designing high V_{OUT} applications.

Selecting the External MOSFETs

The MP2984 requires four external N-channel power MOSFETs (see Figure 12). SWA and SWD are the top two MOSFETs. SWB and SWC are the bottom two MOSFETs. In buck mode, SWA and SWB are switching, and SWD is on. In boost mode, SWC and SWD are switching, and SWA is on.

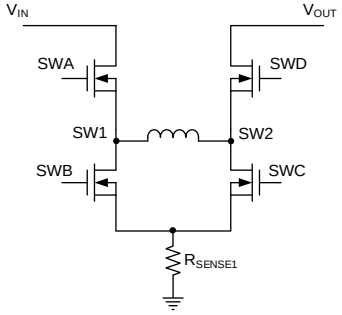


Figure 12: Buck-Boost Topology

The critical parameters for selecting a MOSFET are listed below:

- Maximum drain-to-source voltage (V_{DS_MAX}): SWA and SWB have to withstand the maximum V_{IN} and transient spikes at SW1. It is recommended to set the SWA and SWB V_{DS_MAX} to 1.5 x V_{IN}. SWC and SWD have to withstand V_{OUT} and transient spikes at SW2. It is recommended to set the SWC and SWD V_{DS_MAX} to ≥1.5 x V_{OUT}.
- Maximum continuous drain current (I_{D_MAX}): SWA, SWB, SWC, and SWD have to withstand the maximum inductor current (I_{L_MAX}) and the peak inductor current (I_{L_PEAK}) while switching.
- Voltage threshold (V_{TH}): The gate driver voltages are supplied by VCC. The MOSFET gate plateau voltages should not exceed the minimum V_{CC}; otherwise, the MOSFETs may not enhance fully during start-up or OL conditions.
- Total gate charge (Q_G): All MOSFETs Q_G should be below 50nC (at 7.2V gate condition). If two MOSFETs in parallel, then each MOSFETs' Q_G should be below 25nC.
- Drain-source on resistance ($R_{DS(ON)}$): A lower $R_{DS(ON)}$ improves a MOSFET's temperature rise management and increases efficiency. See the MOSFET descriptions and Equations 16–24 for $R_{DS(ON)}$ selection.

SWA MOSFET

In boost mode, SWA is on, and the SWA conduction power loss ($P_{CON_LOSS_SWA}$) can be calculated with Equation (16):

$$P_{CON_LOSS_SWA} = (I_{OUT}\times\frac{V_{OUT}}{V_{IN}})^2\times R_{DS(ON)_SWA)} \quad (16)$$

If the MOSFET's junction-to-ambient thermal resistance is 50°C/W (determined by the board's power dissipation), and the maximum acceptant temperature rise is 50°C, then the maximum power loss (P_{CON_LOSS}) is 1W. $P_{CON_LOSS_SWA}$ can be calculated with Equation (17):

$$P_{CON_LOSS_SWA} < 1W \quad (17)$$

In buck mode, $P_{CON_LOSS_SWA}$ can be calculated with Equation (18):

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$$P_{CON_LOSS_SWA} = \frac{V_{OUT}}{V_{IN}} \times I_{OUT}^{2} \times R_{DS(ON)_SWA} \quad (18)$$

The SWA switching loss ($P_{SW_LOSS_SWA}$) can be calculated with Equation (19):

$$P_{SW_LOSS_SWA} = \frac{1}{2} V_{IN} \times I_{OUT} \times (t_{ON} + t_{OFF}) \times f_{sw} \quad (19)$$

The MOSFET on time (t_{ON}) and off time (t_{OFF}) are based on Figure 13.

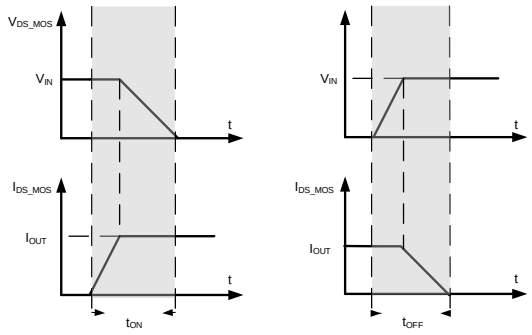


Figure 13: Switch On Time and Switch Off Time

SWB MOSFET

In buck mode, the SWB conduction loss ($P_{CON_LOSS_SWB}$) can be calculated with Equation (20):

$$P_{CON_LOSS_SWB} = (1 - \frac{V_{OUT}}{V_{IN}}) \times I_{OUT}^{2} \times R_{DS(ON)_SWB} \quad (20)$$

SWC MOSFET

In boost mode, SWB is off, and the SWC conduction loss ($P_{CON_LOSS_SWC}$) can be calculated with Equation (21):

$$P_{CON_LOSS_SWC} = (1 - \frac{V_{IN}}{V_{OUT}}) \times (I_{OUT} \times \frac{V_{OUT}}{V_{IN}})^{2} \times R_{DS(ON)_SWC} \quad (21)$$

In boost mode, the SWC switching loss ($P_{SW_LOSS_SWC}$) can be calculated with Equation (22):

$$P_{SW_LOSS_SWC} = \frac{1}{2} \times V_{OUT} \times (I_{OUT} \times \frac{V_{OUT}}{V_{IN}}) \times (t_{ON} + t_{OFF}) \times f_{sw} \quad (22)$$

SWD MOSFET

In buck mode, SWD is on, and the SWD power loss ($P_{CON_LOSS_SWD}$) can be calculated with Equation (23):

$$P_{CON_LOSS_SWD} = I_{OUT}^{2} \times R_{DS(ON)_SWD} \quad (23)$$

In boost mode, the SWD conduction loss can be calculated with Equation (24):

$$P_{CON_LOSS_SWD} = (\frac{V_{IN}}{V_{OUT}}) \times (I_{OUT} \times \frac{V_{OUT}}{V_{IN}})^{2} \times R_{DS(ON)_SWD} \quad (24)$$

The dead time (t_{DEAD}) and the LS-FET switching loss can be ignored.

Compensation Components

The COMP pin controls the system stability and transient response. COMP is the output of the internal error amplifier (EA). A capacitor and resistor in series sets a pole and a zero to control the control the system's characteristics.

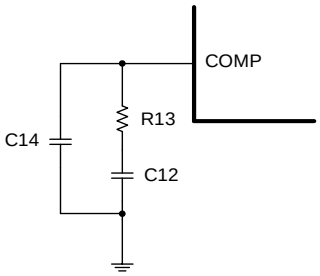


Figure 14: COMP External Compensation

External compensation sets one pole (F_{P1}) and one zero (F_{Z1}) (see Figure 14). F_{P1} can be calculated with Equation (25):

$$F_{P1} = \frac{1}{2\pi \times C14 \times R13} \quad (25)$$

F_{Z1} can be calculated with Equation (26):

$$F_{Z1} = \frac{1}{2\pi \times C12 \times R13} \quad (26)$$

In buck mode, the voltage feedback loop DC gain (A_{VDC}) can be calculated with Equation (27):

$$A_{VDC} = R_{LOAD} \times \frac{G_{CS}}{R_{SENSE1}} \times A_{V-EA} \times \frac{V_{FB}}{V_{OUT}} \quad (27)$$

Where G_{V_EA} is the EA voltage gain (300V/V), G_{CS} is the COMP to current-sense gain, R_{SENSE1} is the current-sense resistor, and R_{LOAD} is the load resistance.

The system has two important poles: one is from the compensation capacitor (C10) and the EA output resistor, and the other one is from C_{OUT} and the load resistor. The compensation capacitor pole (F_{P2}) can be calculated with Equation (28):

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$$F_{P2} = \frac{G_{EA}}{2\pi \times C12 \times G_{V_EA}} \quad (28)$$

The EA output resistor pole (F_{P3}) can be calculated with Equation (29):

$$F_{P3} = \frac{1}{2\pi \times C_{OUT} \times R_{LOAD}} \quad (29)$$

Where G_{EA} is the EA transconductance (1220μA/V).

The system may have another significant zero (F_{ESR}) if C_{OUT} has a large capacitance or a high ESR value. F_{ESR} can be calculated with Equation (30):

$$F_{ESR} = \frac{1}{2\pi \times C_{OUT} \times R_{ESR}} \quad (30)$$

In boost mode, A_{VDC} can be calculated with Equation (31):

$$A_{VDC} = \frac{V_{IN} \times A_{V-EA} \times R_{LOAD} \times V_{FB} \times G_{CS} \times R13}{2 \times V_{OUT}^2 \times R_{SENSE1}} \quad (31)$$

There is also a right half-plane zero (F_{RHPZ}) in boost mode. F_{RHPZ} can be calculated with Equation (32):

$$F_{RHPZ} = \frac{R_{LOAD}}{2 \times \pi \times L} \times (\frac{V_{IN}}{V_{OUT}})^2 \quad (32)$$

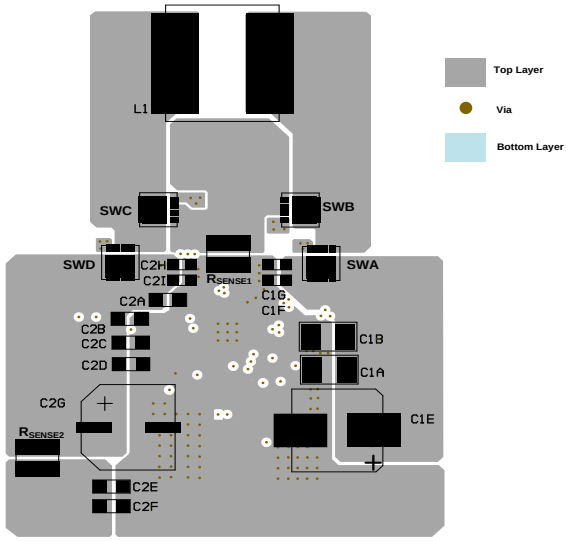
The right half-plane zero increases the gain and reduces the phase, which results in a smaller phase and gain margin. The worst-case condition occurs when V_{IN} is at its minimum and the V_{OUT} is at its maximum.

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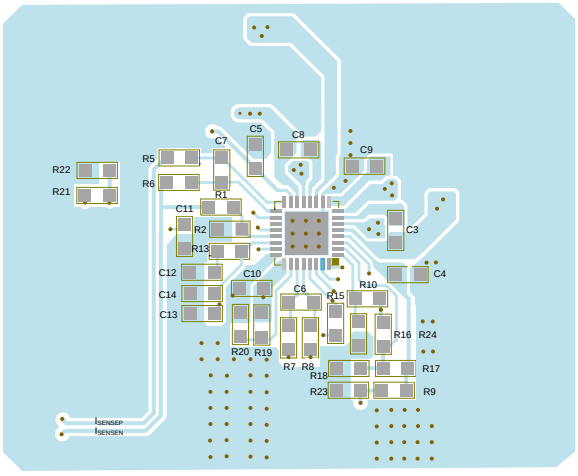
PCB Layout Guidelines

Efficient layout is a critical for stable operation. Improper layout can result in reduced performance, increased EMI, resistive loss, and system instability. For the best results, refer to Figure 15 and follow the guidelines below:

1. In buck mode, place the input power loop (C_{IN}, SWA, and SWB), and the cycle-by-cycle current-sense resistor (R_{SENSE1}) as close to each other as possible.
2. In boost mode, place the output power loop components — including the output filter capacitor (C_{OUT}), the power MOSFETs (SWC and SWD), and the cycle-by-cycle current sense resistor (R_{SENSE1}) — as close as possible.
3. Use wide copper traces and power loop vias to improve thermal dissipation.
4. Connect the exposed pad to GND, and place vias on the exposed pad for thermal dissipation.
5. Place small decoupling capacitors close to VIN, VOUT, and AGND.
6. Make the gate driver traces and return paths as direct as possible.
7. Keep the forward and return traces close together to minimize the inductance of the gate driver path. This can be accomplished by either running the traces side by side or on top of one another on adjacent layers.
8. Use Kelvin connections for R_{SENSE2} (for the average current sense) and R4 (for the cycle-by-cycle current). Run these lines in parallel from the R_{SENSE2}/R_{SENSE1} terminals to the IC pins. Avoid crossing noisy areas, such as SW1, SW2, or gate driver traces.
9. Place the filter capacitor for the current-sense signal as close to the IC's pins as possible.
10. Place the VCC and AVDD capacitors as close as possible to the VCC and AVDD pins.
11. Place the BST1 bootstrap capacitor close to the IC, and connect the capacitor directly to the BST1 and SW1 pins.
12. Place the BST2 bootstrap capacitor close to the IC, and connect the capacitor directly to the BST2 and SW2 pins.
13. Route the feedback loop far away from any noise sources. Place the FB dividers (R1 and R2) as close as possible to the FB and AGND pins.
14. Separate the power and signal paths so that no power or switching current flows through the AGND connections.
15. Connect the PGND and AGND traces near the PGND pin, the VCC capacitor's PGND connection, or near PGND's connection to R4.



Top Layer



Bottom Layer

Figure 15: Recommended PCB Layout

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TYPICAL APPLICATION CIRCUIT

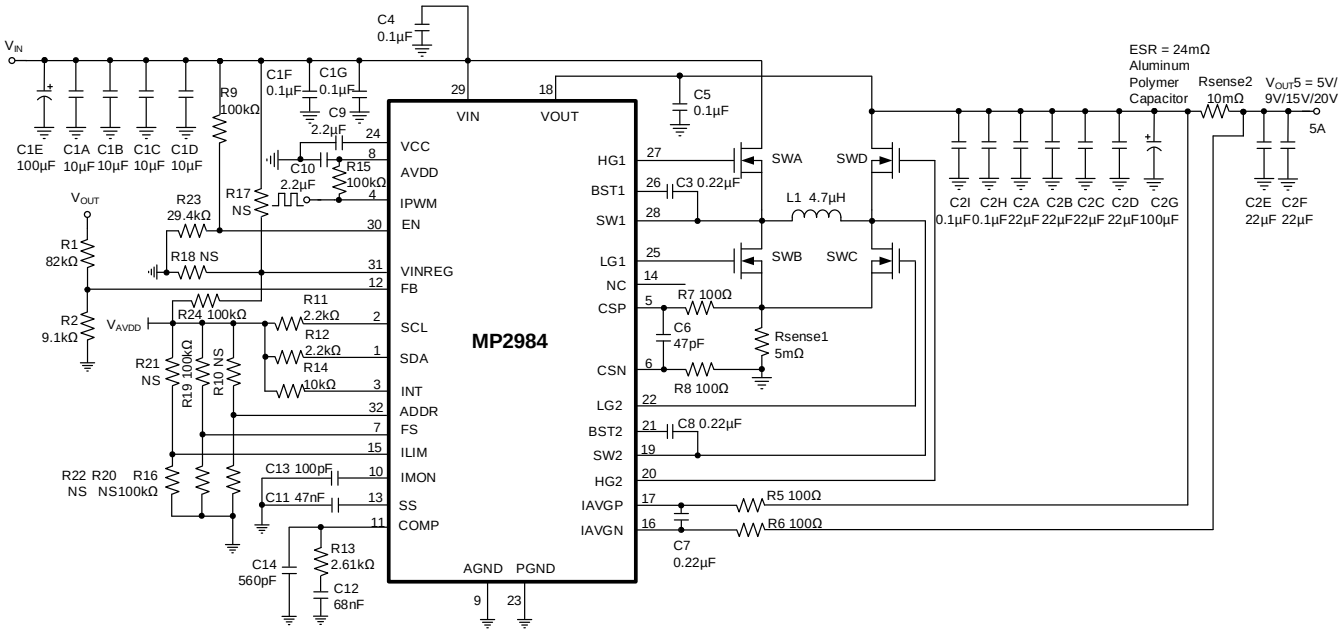
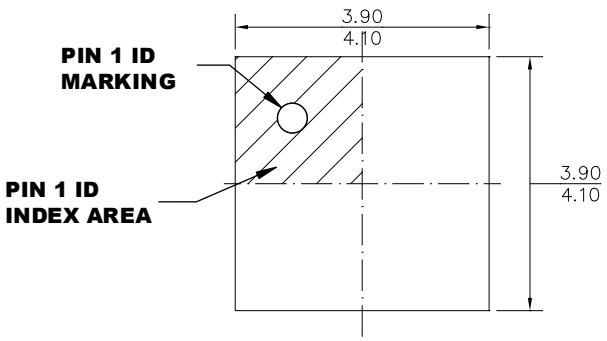


Figure 16: Typical Application Circuit (V_{IN} = 12V, V_{OUT} = 5V/9V/15V/20V for 100W USB PD)

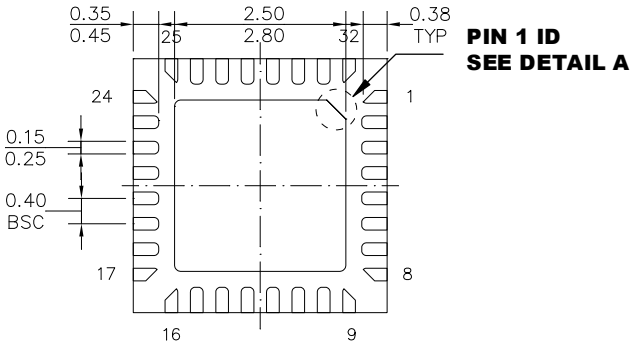
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PACKAGE INFORMATION

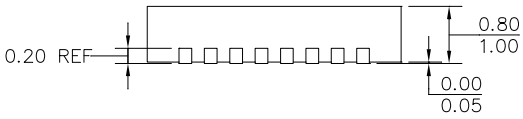
QFN-32 (4mmx4mm)



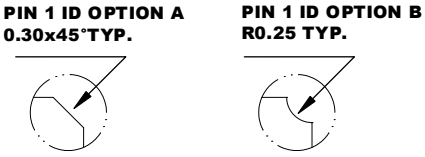
TOP VIEW



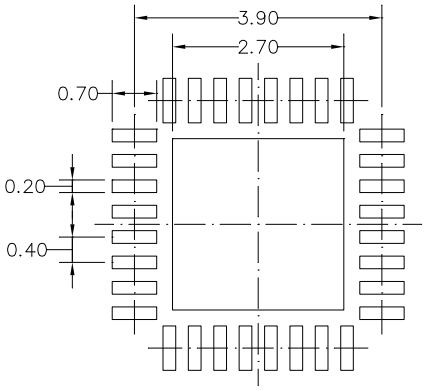
BOTTOM VIEW



SIDE VIEW



DETAIL A



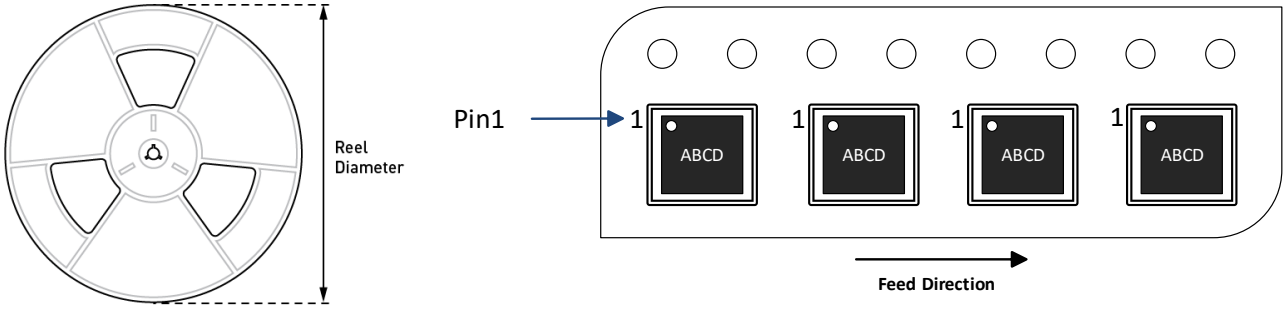
RECOMMENDED LAND PATTERN

NOTE:

1) ALL DIMENSIONS ARE IN MILLIMETERS.
2) EXPOSED PADDLE SIZE DOES NOT INCLUDE MOLD FLASH.
3) LEAD COPLANARITIES SHALL BE 0.1 MILLIMETERS MAX.
4) JEDEC REFERENCE IS MO-220.
5) DRAWING IS NOT TO SCALE.

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CARRIER INFORMATION



Part Number	Package Description	Quantity/ Reel	Quantity/ Tube	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MP2984GR-Z	QFN-32 (4mmx4mm)	5000	N/A	13in	12mm	8mm

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REVISION HISTORY

Revision #	Revision Date	Description	Pages Updated
1.0	12/09/2021	Initial Release	-

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